A Theory of Information Architecture for Networked Decisions: Freshness, Locality, and Coordination

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Abstract

Networked decisions often face a tradeoff between information scope and freshness: broader information can improve coordination but may arrive with greater delay. We study this tradeoff for repeated quadratic static-team decisions in an exogenously evolving stochastic environment. An information architecture specifies what each agent knows, from where, and at what age, and its performance is measured by the loss relative to the current full-information decision. First, we compare the pure fresh-local and stale-global benchmarks. The crossover is governed by decision-relevant temporal predictability and the share of decision-relevant variation captured by local information; under a common-rate Gauss–Markov model with affine full-information decisions, this yields a closed-form threshold in the dimensionless delay τ/T. Second, for a synchronized-snapshot radius family in which broader neighborhood snapshots arrive at greater age, we derive an exact composition law between spatial omission and temporal staleness and characterize the optimal radius within that family by balancing marginal spatial-information gain against marginal freshness loss. In a canonical field model, the resulting scaling depends on the spatial correlation length ℓ_s, decision-relevance length ℓ_c, and temporal propagation length $L_T = vT$. Numerical experiments verify the predicted crossover, zero-to-positive-radius threshold, interior optimal-radius behavior, and corresponding shifts on finite noncanonical graphs. Within the analyzed exogenous-state quadratic framework, the resulting laws are governed by predictability of the full-information decision rather than raw state-prediction accuracy alone.

1 Introduction

Many networked systems must repeatedly make coupled decisions using information that is distributed across space and changes over time. A wireless transmitter may know its current local channel state but not the current interference conditions elsewhere in the network; an edge server may observe its own workload immediately while a system-wide view is available only after telemetry has been collected and disseminated; and a distributed sensing system may have fresh measurements from nearby sensors while observations from distant sensors arrive with greater latency. In each case, broad information can improve coordination, but obtaining it takes time. The resulting design choice is not simply between centralized and distributed control, but between *information architectures* that differ simultaneously in spatial scope and temporal freshness.

This tension appears in several literatures, sometimes with scope and delay treated jointly but under substantially different performance models. Work on stale information in distributed systems shows that old global state can lose much of its value [1]; Age of Information formalizes freshness as a system resource [2]; distributed optimization studies computation under communication constraints [3]; organizational economics studies adaptation to local information versus coordination [4]; and networked-control work directly examines how communication scope, topology, and propagation delay interact with closed-loop performance.

The closest control-theoretic precedents make the overlap particularly clear. Distributed-control work on spatially invariant systems establishes controller localization and imposes finite-speed or distance-dependent communication constraints through cone- and funnel-causal formulations and state-space models of finite communication speed [5–8]. More directly, Ballotta, Jovanović, and Schenato parameterize a controller architecture by the number of communication hops available to each agent and assign the measurements in that scope an increasing architecture-dependent delay; after optimizing the feedback gains for each architecture, they show that sparse feedback can outperform all-to-all control [9]. Ballotta and Gupta obtain a related sparse optimum for consensus convergence speed [10], while Vanka, Gupta, and Haenggi study the connectivity–delay tradeoff induced by wireless communication radius and interference [11]. Subsequent work studies how spatially varying measurement delays reshape the locality of optimal feedback in spatially invariant plants [12]. Thus neither scope-dependent communication delay nor the possibility that a sparse architecture outperforms a dense one is new in itself.

The distinction pursued here is the decision problem and the resulting analytical object. The environment evolves exogenously, and each epoch is a static team decision rather than a dynamic feedback-control problem in which the controller changes future plant state. For a fixed information architecture, performance is the Q-weighted loss in reproducing the current full-information decision. We then vary spatial scope and information age jointly. Under the common-rate model, this prediction-theoretic formulation yields an exact composition identity for spatial omission and temporal staleness, a fresh-local/stale-global crossover in terms of decision-relevant predictability, and a marginal-balance characterization of the synchronized-snapshot radius. In the canonical field model, these quantities produce an explicit radius law involving temporal coherence, spatial correlation, decision-relevance length, and propagation speed.

This paper develops a theory for this tradeoff. We consider a collection of agents operating in an exogenously evolving stochastic environment $\theta_t = (\theta_{1,t}, \ldots, \theta_{n,t})$. At each decision epoch the agents select a joint action $x_t = (x_{1,t}, \ldots, x_{n,t})$ that incurs a common cost $f(x_t, \theta_t)$. An *information architecture* specifies the information available to each agent when this decision is made. Thus a fresh-local architecture may allow agent i to condition on the current $\theta_{i,t}$, a neighborhood architecture may provide the states of agents within a given spatial radius, and a global architecture may provide the entire state vector only after a delay τ. More generally, an architecture specifies *who knows what, from where, and at what age.*

Our analysis uses the classical fixed-information framework of static team decision theory initiated by Radner [13]. In the deterministic-Hessian quadratic model considered here, completing the square reduces optimization under any fixed information architecture to a Hilbert-space approximation problem: the team-optimal policy is the Q-orthogonal projection of the full-information action onto the closed subspace of implementable policies. Consequently, the architecture regret is

$$R(\mathcal{A}) = \frac{1}{2} \left\| x_t^\star - \Pi_{\mathcal{S}(\mathcal{A})} x_t^\star \right\|_Q^2 , \tag{1}$$

where $\mathcal{S}(\mathcal{A})$ denotes the admissible policy subspace. We use the fixed-architecture static-team solution as the inner optimization primitive and study a particular outer design problem: comparison and selection within a structured family in which information scope and age vary jointly. Classical team and information-structure work has also treated information and organizational networks as design objects [14,15]. The narrower question pursued here is how scope-dependent age changes the value of a synchronized-snapshot radius family when information is evaluated through prediction error of the full-information decision.

Classical team theory established fixed-information optimality conditions and also treated the value and organization of information, while later work has designed measurement or communication structures and studied radius-limited local information [14–16]. The narrower contribution here is an exact prediction-theoretic characterization of a structured scope–age family in an exogenous-state quadratic static team, including the pure-benchmark crossover, the common-rate spatio-temporal composition identity, and the resulting radius scaling laws.

Our first result answers the two extreme points of this tradeoff. We compare a *fresh-local* architecture, in which each agent observes its current local state, against a *stale-global* architecture, in which every agent has access to the complete system state with delay τ. In general, the loss from staleness is governed by the prediction error of the current decision-relevant state from information τ units old. For a Gauss–Markov environment with temporal coherence time T, and for an affine full-information decision, the global architecture has the exact regret

$$R_{\mathrm{glob}}(\tau) = \left(1 - e^{-2\tau/T}\right) R_\infty, \tag{2}$$

where R_∞ is the regret of the best open-loop decision. If R_{loc} denotes fresh-local regret and

$$\eta_{\mathrm{loc}} \triangleq 1 - \frac{R_{\mathrm{loc}}}{R_\infty} \tag{3}$$

is the fraction of decision-relevant variation explainable from fresh-local information, then fresh-local decisions outperform stale-global decisions exactly when

$$\frac{\tau}{T} > \frac{1}{2} \log \frac{1}{\eta_{\mathrm{loc}}}. \tag{4}$$

Thus the value of global coordination is controlled by two independent quantities: how rapidly global information loses predictive value in time, and how much of the globally optimal decision is already predictable locally. For a canonical shared-resource quadratic model with local stiffness q and coordination strength γ, the large-system threshold reduces to

$$\left(\frac{\tau}{T}\right)^{\star} = \frac{1}{2} \log \left(1 + \frac{\gamma}{q}\right), \tag{5}$$

making explicit how stronger decision coupling increases the amount of staleness that is worth tolerating in exchange for global coordination.

The local–global comparison concerns two pure benchmark regimes. Our second result considers agents embedded in a graph or spatial domain and a synchronized-snapshot family in which each agent bases its decision on a time-aligned neighborhood snapshot of radius r. Increasing r replaces a narrower, fresher snapshot by a broader snapshot of greater age $\tau(r)$. The resulting architecture faces two competing sources of error: a *spatial omission error*, caused by excluding decision-relevant state beyond radius r, and a *temporal staleness error*, caused by the delay required to acquire the broader information.

We characterize this tradeoff using the spatial and temporal predictability of the underlying stochastic field. In a canonical model, spatial dependence is described by a correlation length ℓ_s, temporal variation by a coherence time T, and communication by the scope–delay relation $\tau(r)$. We show that the optimal radius within the synchronized-snapshot family

$$r^\star \in \arg\min_r R\big(\mathcal{A}_{r,\tau(r)}\big) \tag{6}$$

is determined jointly by these environmental scales and by the coupling structure of the decision objective. The result yields comparative statics with direct architectural interpretations: faster temporal variation favors smaller and fresher information radii; faster information propagation supports broader coordination; and stronger long-range decision coupling increases the value of larger information scope. Spatial smoothness determines how rapidly additional spatial observations become redundant and hence how much information radius is needed to approximate global decisions. Under tractable spatio-temporal covariance and propagation models, these effects combine into dimensionless scaling laws for $r^\star$.

The distinction between *state predictability* and *decision-relevant predictability* is important. A remote state variable may be difficult to predict yet irrelevant to the optimal action, while a small uncertainty in another variable may matter greatly because of strong coupling in the objective. Our results therefore weight spatio-temporal uncertainty by its effect on the full-information decision. This makes the appropriate coordination scale a property not only of the stochastic environment, but of the combination of the environment, the decision problem, and the communication architecture.

The framework deliberately separates information architecture from the algorithm used to implement it. It consequently complements canonical distributed-optimization formulations, where the communication structure is specified first and the principal question is how to compute efficiently over it [3]. We study a structured outer comparison over a scope–age family. Similarly, unlike freshness metrics that assign value to an update primarily through its age [2], the value of an observation here depends on how much it changes the downstream decision. This perspective also suggests extensions to heterogeneous information radii, sparse long-range information "shortcuts," low-dimensional global summaries, and budget-constrained information acquisition.

Our model is intentionally restricted to *static* information structures: the stochastic environment may be temporally and spatially correlated, but agents' current actions do not alter the information subsequently observed by other agents. This boundary is consequential. Once actions influence future observations, they may serve simultaneously as control actions and implicit signals, leading to the nonclassical dynamic-team phenomena highlighted by Witsenhausen's counterexample [17]. Thus the present theory is most directly applicable to repeated networked optimization against an exogenously evolving environment—for example, stylized quadratic or locally quadratic models of rate, power, compute, or sensing decisions driven by changing demand, channel, or resource conditions—rather than to fully constrained engineering formulations, controlled queueing, or decentralized Markov decision processes.

In summary, this paper makes two technical contributions and one conceptual contribution:

1. We derive an exact *fresh-local versus stale-global crossover law* for two pure information regimes, expressed through decision-relevant temporal predictability and the share of decision variation captured locally.

2. For a synchronized-snapshot radius family under the common-rate model, we derive an exact spatio-temporal prediction-error composition law and use it to characterize the optimal radius by a marginal balance between spatial-information gain and freshness loss. In a canonical field model, this yields an explicit scaling law in ℓ_s, ℓ_c, and vT.

3. We show that, within the analyzed framework, architecture performance is governed by *decision-relevant predictability*, not raw state predictability. The desired architecture depends jointly on environmental statistics, decision sensitivity and coupling, information geometry, and freshness; the communication graph and decision geometry therefore need not coincide.

Together, these results support normalized comparisons within a structured family that goes beyond a binary choice between centralized and distributed decision making. Within that family, the preferred radius is the spatial scale at which the marginal value of broader coordination is balanced by the loss of information freshness required to obtain it.

2 Networked Decisions and Information Architectures

We consider a network of decision-making agents operating in a stochastic environment whose state is distributed across the network and evolves over time. Our formulation separates three distinct objects: the *network geometry*, which defines relationships and distances among agents; the *decision problem*, which determines how their actions and states interact in the objective; and the *information architecture*, which determines what information is available to each agent and with what delay. These three structures need not coincide.

The environment evolves exogenously: agents react to the state but their current actions do not alter the law governing its future evolution. The model therefore describes repeated information-constrained decisions rather than a controlled Markov process. Under the fixed-information static-team formulation, the present deterministic-Hessian quadratic model admits the projection characterization developed below. We use that fixed-architecture result as the inner optimization primitive for comparing structured scope–age architectures.

2.1 Network, Environment, and Decision Model

There are n decision-making agents indexed by

$$V = \{1, \dots, n\},$$

represented as the vertices of a graph

$$G = (V, E). \tag{7}$$

The graph provides a notion of network or spatial proximity. We write $d_G(i, j)$ for the graph distance between nodes i and j, and define the radius-r neighborhood of node i as

$$\mathcal{N}_r(i) \triangleq \{j \in V : d_G(i, j) \le r\}. \tag{8}$$

Depending on the application, G may represent a physical communication network, a geographic neighborhood relation, or another structure governing the cost or latency of obtaining nonlocal information. [1]

Let

$$\theta = \{\theta_t\}_{t \in \mathbb{T}}$$

be a stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathbb{T}$ may be discrete or continuous. At time t, the environment state is

$$\theta_t = (\theta_{1,t}, \dots, \theta_{n,t}) \in \mathbb{R}^m, \qquad \theta_{i,t} \in \mathbb{R}^{m_i}, \qquad \sum_{i=1}^{n} m_i = m. \tag{9}$$

The component $\theta_{i,t}$ denotes the state locally associated with node i. Depending on the application, it may represent a local channel condition, workload or demand, sensing quality, resource availability, or another externally evolving quantity relevant to the decision.

For the basic results in this section, we assume that θ_t is stationary and square-integrable:

$$\mathbb{E}\|\theta_t\|^2 < \infty. \tag{10}$$

[1] The graph model is adopted for concreteness. The framework can more generally be formulated for agents embedded in a metric space, or for any collection of agents equipped with a meaningful notion of neighborhoods and information-propagation cost.

No independence, Gaussianity, Markov property, or specific form of spatial or temporal correlation is required. More structured stochastic models will be introduced later when we derive explicit architecture laws.

Agent i chooses an action

$$x_{i,t} \in \mathbb{R}^{d_i},$$

and the joint action is

$$x_t = (x_{1,t}, \dots, x_{n,t}) \in \mathbb{R}^d, \qquad d = \sum_{i=1}^{n} d_i. \tag{11}$$

The agents share a common per-stage cost

$$f(x, \theta) = \frac{1}{2} x^\top Q x - b(\theta)^\top x + c(\theta), \tag{12}$$

where $Q \in \mathbb{R}^{d \times d}$ is symmetric and positive definite, $b : \mathbb{R}^m \to \mathbb{R}^d$ is measurable, and $c : \mathbb{R}^m \to \mathbb{R}$ is an additive state-dependent term. We assume

$$\mathbb{E}\|b(\theta_t)\|^2 < \infty, \qquad \mathbb{E}|c(\theta_t)| < \infty. \tag{13}$$

The matrix Q describes coupling among the agents' actions. If Q is partitioned according to $x = (x_1, \dots, x_n)$, an off-diagonal block Q_{ij} captures direct interaction between the decisions of agents i and j. The mapping $b(\theta)$ provides another form of coupling: the preferred action of agent i may depend on states associated with other agents.

We emphasize that this decision coupling is distinct from the graph G. The graph describes network proximity and information propagation, whereas Q and $b(\cdot)$ determine which states and actions are relevant to one another in the objective. Nearby agents need not be strongly coupled, and strong decision coupling may exist between distant nodes.

If the complete current state θ_t is available without information constraints, the unique optimal joint action is

$$x_t^\star \triangleq x^\star(\theta_t) = Q^{-1} b(\theta_t). \tag{14}$$

We call $x_t^\star$ the *clairvoyant* or *full-information* decision.

For the affine specializations used to obtain closed forms, we write

$$b(\theta) = B\theta + b_0, \qquad B \in \mathbb{R}^{d \times m}, \qquad b_0 \in \mathbb{R}^d, \qquad K \triangleq Q^{-1} B \in \mathbb{R}^{d \times m}. \tag{15}$$

The matrix B maps state into the linear term of the objective, while K is the resulting decision-sensitivity operator. The general projection results do not require this affine form.

Completing the square in (12) gives

$$f(x, \theta) - f(x^\star(\theta), \theta) = \frac{1}{2} \left(x - x^\star(\theta)\right)^\top Q \left(x - x^\star(\theta)\right). \tag{16}$$

Hence, for the quadratic model, excess decision cost is exactly a weighted squared error in reproducing the full-information action.

2.2 Information Architectures

An *information architecture* specifies what information is available to each agent when its decision is made. At a fixed decision epoch, let

$$\mathcal{G}_i \subseteq \mathcal{F}$$

denote the σ-algebra representing the information available to agent i. An information architecture is the tuple

$$\mathcal{A} = (\mathcal{G}_1, \ldots, \mathcal{G}_n). \tag{17}$$

A decision rule for agent i is admissible only if it is measurable with respect to $\mathcal{G}_i$.

This abstraction accommodates information patterns that differ in spatial scope, temporal freshness, and degree of aggregation.

Fresh-local information. Agent i observes its own current state and local history:

$$\mathcal{G}_{i,t}^{\mathrm{loc}} = \sigma(\theta_{i,s} : s \leq t)\,. \tag{18}$$

This architecture has minimal spatial scope and no imposed information delay.

Stale-global information. Every agent has access to the complete state history, but only up to time $t - \tau$:

$$\mathcal{G}_{i,t}^{\mathrm{glob}}(\tau) = \sigma(\theta_s : s \leq t - \tau)\,, \qquad i = 1, \ldots, n. \tag{19}$$

This architecture has maximal spatial scope but potentially substantial staleness.

Radius-r information. More generally, agent i may obtain state information from nodes within graph distance r:

$$\mathcal{G}_{i,t}^{(r)} = \sigma\big(\theta_{j,t-\tau(r)} : j \in \mathcal{N}_r(i)\big)\,. \tag{20}$$

The radius family studied below is a synchronized-snapshot architecture. At radius r, the decision is formed from a time-aligned neighborhood snapshot whose age is $\tau(r)$; fresher local observations are not combined with the delayed snapshot. This restriction models applications requiring temporal consistency or a common decision epoch. If fresh local and delayed remote information can instead be freely combined, the resulting hybrid architecture contains more information and weakly dominates either source alone; such mixed-age architectures are outside the principal family analyzed here.

A closely related common-delay-by-scope abstraction appears in Ballotta, Jovanović, and Schenato, where each agent receives measurements from nodes within a selected communication-hop radius and the measurements in that scope share an architecture-dependent delay that increases with the radius [9]. Their setting is dynamic feedback control rather than the exogenous-state static-team setting considered here, and their information structures are not identical to (20).

Here $\tau(r)$ denotes the latency required to acquire information over radius r. We are particularly interested in systems for which

$$r_2 \geq r_1 \quad \Longrightarrow \quad \tau(r_2) \geq \tau(r_1), \tag{21}$$

so that broader information is systematically less fresh.

The synchronized-snapshot radius family interpolates between its local and global endpoints under the stated temporal model. At the smallest radius, each decision relies on a narrow, rapidly available snapshot. As r increases, that snapshot is replaced by a broader but older view. When r reaches the diameter of a finite graph, (20) is a delayed global snapshot, not the complete delayed global history in (19).

Local information plus a global summary. An architecture need not transport raw remote state. Agent i may instead combine fresh-local information with a low-dimensional statistic

$$z_t = h(\theta_t)$$

of the global state:

$$\mathcal{G}_{i,t}^{\mathrm{hyb}} = \sigma(\theta_{i,s} : s \le t) \vee \sigma(z_s : s \le t - \tau_z)\,. \tag{22}$$

Examples include congestion prices, aggregate load indicators, or other summaries intended to communicate the globally decision-relevant portion of the state. This hybrid automatically contains the fresh-local information set. It contains, and hence weakly dominates, the stale-global benchmark only when the delayed summary preserves the complete delayed global information; an arbitrary low-dimensional summary need not dominate that benchmark.

For a given architecture $\mathcal{A}$, define the admissible joint-policy set

$$\mathcal{S}(\mathcal{A}) \triangleq \big\{u = (u_1, \ldots, u_n) : u_i \in L^2,\ u_i \text{ is } \mathcal{G}_i\text{-measurable},\ i = 1, \ldots, n\big\}\,. \tag{23}$$

Thus an information architecture constrains the information on which each decision may depend; it does not prescribe a specific algorithm for computing that decision.

The information structures considered here are *static* in the team-decision sense. The sets $\mathcal{G}_i$ are exogenously specified: current actions neither change the future evolution law of θ_t nor alter the information available to other agents. This excludes action-based signaling and controlled state evolution, and distinguishes the present model from dynamic team and decentralized control problems.

2.3 Architecture Regret

We evaluate an architecture by the best expected performance achievable under its information restrictions. Define

$$J(\mathcal{A}) \triangleq \inf_{u \in \mathcal{S}(\mathcal{A})} \mathbb{E}\left[f(u, \theta_t)\right], \tag{24}$$

and let

$$J^\star \triangleq \mathbb{E}\left[f(x_t^\star, \theta_t)\right] \tag{25}$$

denote the expected full-information cost.

The *architecture regret* is

$$R(\mathcal{A}) \triangleq J(\mathcal{A}) - J^\star. \tag{26}$$

Here "regret" denotes a static expected performance gap after optimizing over all policies admissible under the architecture. It is not cumulative regret in a sequential-decision or online-learning sense.

Because the environment is stationary, this expected per-stage regret is independent of the particular decision epoch, and the expected finite-horizon average equals the same one-stage expectation. Almost-sure convergence of sample-path time averages additionally requires ergodicity.

At this stage, $R(\mathcal{A})$ measures only the decision-quality consequence of an information restriction. We do not separately charge for communication, sensing, or computation. The physical cost of obtaining broader information enters through the feasible architecture and, in particular, through the scope–latency relation $\tau(r)$.

If two architectures $\mathcal{A}$ and $\mathcal{A}'$ satisfy

$$\mathcal{G}_i \subseteq \mathcal{G}_i', \qquad i = 1, \ldots, n, \tag{27}$$

then every policy feasible under $\mathcal{A}$ is also feasible under $\mathcal{A}'$, and therefore

$$R(\mathcal{A}') \le R(\mathcal{A}). \tag{28}$$

Thus additional information cannot hurt when it is available with the same freshness. The interesting architectural tradeoff arises because increasing spatial scope generally also increases information age.

2.4 Fixed-Architecture Quadratic Projection

The classical fixed-information static-team formulation supplies coupled person-by-person optimality conditions [13]. In the present deterministic-Hessian quadratic model, completing the square turns those conditions into the especially clean projection characterization below. We use this fixed-architecture specialization as the analytical primitive for the outer scope–age comparison. Related affine-gain projection formulations appear in Krainak, Speyer, and Marcus [18].

Let

$$\mathcal{H} = L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d) \tag{29}$$

and equip $\mathcal{H}$ with the weighted inner product

$$\langle u, v\rangle_Q \triangleq \mathbb{E}\left[u^\top Q v\right], \qquad \|u\|_Q^2 \triangleq \langle u, u\rangle_Q. \tag{30}$$

Since $Q \succ 0$, the induced norm is equivalent to the ordinary L^2 norm.

Lemma 1 (Fixed-architecture quadratic projection; Radner-type specialization)**.** *For any static information architecture $\mathcal{A}$, the admissible policy set $\mathcal{S}(\mathcal{A})$ defined in* (23) *is a closed linear subspace of $\mathcal{H}$. Under the quadratic cost* (12)*, the unique optimal policy implementable under $\mathcal{A}$ is*

$$u^\star_{\mathcal{A}} = \Pi_{\mathcal{S}(\mathcal{A})} x_t^\star, \tag{31}$$

where $\Pi_{\mathcal{S}(\mathcal{A})}$ denotes orthogonal projection with respect to (30)*. Consequently,*

$$\boxed{R(\mathcal{A}) = \frac{1}{2}\left\|x_t^\star - \Pi_{\mathcal{S}(\mathcal{A})} x_t^\star\right\|_Q^2.} \tag{32}$$

This identity is the deterministic-Hessian quadratic specialization of the classical fixed-information static-team formulation; it is not the literal form in which Radner's theorem is usually stated.

Proof. Linearity of $\mathcal{S}(\mathcal{A})$ follows because $\mathcal{G}_i$-measurability is preserved under linear combinations. Closedness follows because an L^2 limit of $\mathcal{G}_i$-measurable random variables admits a $\mathcal{G}_i$-measurable version.

For any admissible policy u, (16) gives

$$\begin{aligned}\mathbb{E}\left[f(u, \theta_t) - f(x_t^\star, \theta_t)\right] &= \frac{1}{2}\mathbb{E}\left[(u - x_t^\star)^\top Q (u - x_t^\star)\right] \\ &= \frac{1}{2}\|u - x_t^\star\|_Q^2.\end{aligned} \tag{33}$$

Minimizing this expression over the closed subspace $\mathcal{S}(\mathcal{A})$ is the standard Hilbert-space projection problem, yielding (31) and (32). $\square$

Corollary 1 (Coupled conditional normal equations)**.** *The unique team optimum in Lemma 1 satisfies*

$$\boxed{\mathbb{E}\big[\big(Qu^\star_{\mathcal{A}} - b(\theta_t)\big)_i \mid \mathcal{G}_i\big] = 0, \qquad i = 1, \ldots, n.} \tag{34}$$

With Q partitioned into action blocks, this is equivalently

$$Q_{ii}u^\star_i + \sum_{j\neq i} Q_{ij}\,\mathbb{E}[u^\star_j \mid \mathcal{G}_i] = \mathbb{E}[b_i(\theta_t) \mid \mathcal{G}_i]. \tag{35}$$

These are the coupled person-by-person stationarity equations associated with the classical static-team formulation. Strict convexity makes them sufficient for the unique team optimum here. With heterogeneous information and off-diagonal Q, one generally does not have $u^\star_i = \mathbb{E}[x^\star_i \mid \mathcal{G}_i]$. When all components share a common information sigma-algebra, deterministic Q makes the joint solution the ordinary vector conditional expectation.

Remark 1 (Scope of the projection identity)**.** *The fixed-architecture projection identity requires square integrability, a deterministic positive-definite matrix Q, unconstrained Euclidean actions, and a closed linear space of admissible policies. It does not require Gaussianity, an affine map $b(\theta)$, Markov dynamics, temporal stationarity, or independent observations. Those stronger assumptions enter only in later closed-form prediction and composition results.*

Lemma 1 turns an information architecture into a geometric object. The full-information policy $x^\star_t$ specifies what the system would ideally do if the current global state were known. The architecture restricts the system to the policy subspace $\mathcal{S}(\mathcal{A})$, and its regret is precisely the squared distance from the ideal policy to the closest rule implementable using the available information.

This interpretation also identifies the relevant notion of environmental predictability. An architecture need not reconstruct the complete state θ_t. It need only provide enough information to predict those components of the state that materially affect $x^\star_t$. We therefore focus throughout on *decision-relevant predictability*, rather than raw state-prediction accuracy.

The radius-dependent architecture in (20) makes the central tradeoff particularly clear. Increasing r provides each agent with a broader view of the network and, if freshness were fixed, could only improve performance. In a physical network, however, larger r generally also increases $\tau(r)$. The architecture family

$$\big\{\mathcal{A}_{r,\tau(r)}\big\}_{r\geq 0} \tag{36}$$

therefore trades spatial information against temporal freshness.

We first study the two endpoints of this tradeoff: fresh-local information and stale-global information. We then ask the more general architecture-design question: what information radius minimizes regret when the value of broader spatial knowledge must be balanced against the additional delay required to obtain it?

The notation admits a direct networked interpretation, but the mappings are illustrative rather than application case studies. The component θ_i may represent an externally evolving local condition—for example, channel quality, demand, workload intensity, sensing conditions, or resource availability—while x_i is a corresponding continuous decision such as rate, power, compute allocation, or sensing effort. The matrices Q and B, or equivalently the decision-sensitivity operator $K = Q^{-1}B$, encode how other agents' decisions and remote state affect the desired action; this coupling structure need not coincide with the physical or information graph over which observations travel. The essential boundary is exogeneity. An external arrival or workload process may be modeled as part of θ, whereas a queue length whose evolution depends on earlier service decisions is not exogenous in the sense required here. Likewise, channel conditions fit the model when the decisions under study do not materially determine the future channel-state law. Thus these interpretations connect the abstract framework to networked resource decisions while preserving the static-team assumption that current actions neither control future environmental state nor reshape another agent's information.

Table 1: Core notation. Quantities labeled as shares are normalized by R_∞.

Symbol	Meaning	Symbol	Meaning
n	number of agents	$G=(V,E)$	network or geometry graph
θ_t	exogenous environment state	x_t	joint action
Q	positive-definite action-coupling matrix	B	affine state-to-objective map
$K=Q^{-1}B$	decision-sensitivity operator	$x_t^\star$	full-information decision
$\mathcal{A}$	information architecture	$R(\mathcal{A})$	architecture regret; includes R_{loc} and $R_{\text{glob}}(\tau)$
R_∞	open-loop regret	η_{loc}	share captured by local information
$\eta_T(\tau)$	temporal unpredictability	$\eta_S(r)$	zero-delay spatial omission
$\eta_{ST}(r,\tau)$	normalized joint spatio-temporal regret	T	temporal coherence time
$\tau(r)$	information latency at radius r	v	effective propagation speed
ℓ_s	spatial correlation length	ℓ_c	decision-relevance length
$L_T=vT$	temporal propagation length	$r^\star$	optimal synchronized-snapshot radius

3 Fresh-Local versus Stale-Global Information

We begin with the two endpoints of the freshness–scope tradeoff introduced in Section 2. A *fresh-local* architecture gives each agent immediate access to its own state but no current information about remote states. A *stale-global* architecture gives every agent complete system-wide information, but only after a delay τ.

The comparison in this section concerns two mutually exclusive pure information regimes. It does not imply that an agent should discard current local information when that information can be freely combined with delayed global history. The union architecture $\mathcal{G}_{i,t}^{\text{loc}} \vee \mathcal{G}_t^{\tau}$ contains both benchmark information sets and therefore weakly dominates each. The crossover below instead compares protocol or organizational regimes in which the operational decision is based on one of the two pure information patterns; mixed-age architectures are outside the present analysis.

If both architectures had the same delay, global information could only help: the corresponding information sets would contain the local ones. The comparison becomes nontrivial precisely because spatial scope and freshness move in opposite directions. This section characterizes that tradeoff first for an arbitrary stationary environment through a temporal prediction-error function, and then obtains an exact crossover law under a Gauss–Markov model.

3.1 The Two Architectures

At decision time t, the *fresh-local* architecture is

$$\mathcal{A}_{\text{loc}} = \left(\mathcal{G}_{1,t}^{\text{loc}}, \dots, \mathcal{G}_{n,t}^{\text{loc}}\right), \qquad \mathcal{G}_{i,t}^{\text{loc}} = \sigma(\theta_{i,s} : s \le t)\,. \tag{37}$$

Its regret is

$$R_{\text{loc}} \triangleq R(\mathcal{A}_{\text{loc}}). \tag{38}$$

Because this information is current, R_{loc} does not depend on the global information delay τ. Its magnitude instead reflects how much of the full-information decision can be inferred from local information alone.

The *stale-global* architecture at delay τ is

$$\mathcal{A}_{\text{glob}}(\tau) = (\mathcal{G}_t^\tau, \dots, \mathcal{G}_t^\tau), \qquad \mathcal{G}_t^\tau \triangleq \sigma(\theta_s : s \leq t - \tau). \tag{39}$$

All agents therefore share the same complete but delayed view of the environment. We denote its regret by

$$R_{\text{glob}}(\tau) \triangleq R(\mathcal{A}_{\text{glob}}(\tau)). \tag{40}$$

For reference, define also the *open-loop* architecture, under which no state-dependent information is available and the joint decision must be deterministic. Its optimal action is $\mathbb{E}[x_t^\star]$, and its regret is

$$R_\infty \triangleq \frac{1}{2}\mathbb{E}\left[\|x_t^\star - \mathbb{E}[x_t^\star]\|_Q^2\right]. \tag{41}$$

Throughout this section we assume $R_\infty > 0$, excluding the trivial case in which the clairvoyant decision is deterministic.

3.2 General Predictability Characterization

The stale-global architecture is particularly simple because all agents share one common information set. Its admissible policy space is therefore

$$L^2(\Omega, \mathcal{G}_t^\tau, \mathbb{P}; \mathbb{R}^d).$$

Since Q is deterministic, the Q-weighted orthogonal projection of $x_t^\star$ onto this space is simply its conditional expectation.

Proposition 1 (Common-information projection and stale-global prediction error)**.** *For any stationary environment satisfying the assumptions of Section 2,*

$$u_{\text{glob}}^\star(\tau) = \mathbb{E}\left[x_t^\star \mid \mathcal{G}_t^\tau\right], \tag{42}$$

and

$$R_{\text{glob}}(\tau) = \frac{1}{2}\mathbb{E}\left[\|x_t^\star - \mathbb{E}\left[x_t^\star \mid \mathcal{G}_t^\tau\right]\|_Q^2\right] \tag{43}$$

$$= \frac{1}{2}\mathbb{E}\left[\operatorname{tr}\left(Q \ \operatorname{Cov}\left(x_t^\star \mid \mathcal{G}_t^\tau\right)\right)\right]. \tag{44}$$

Proof. By Lemma 1, the optimal stale-global policy is the orthogonal projection of $x_t^\star$ onto the set of $\mathcal{G}_t^\tau$-measurable joint actions. For any $\mathcal{G}_t^\tau$-measurable v,

$$\mathbb{E}\left[\left(x_t^\star - \mathbb{E}[x_t^\star \mid \mathcal{G}_t^\tau]\right)^\top Qv\right] = 0,$$

so this projection is the conditional expectation in (42). Substitution into (32) yields (43), and the conditional-covariance form follows from the standard mean-square prediction identity. $\square$

Proposition 1 reveals that the effect of staleness is fundamentally a *prediction* problem. Delay matters only to the extent that information available at time $t - \tau$ fails to predict the decision that would be optimal at time t.

This motivates the normalized *temporal unpredictability function*

$$\eta_T(\tau) \triangleq \frac{R_{\mathrm{glob}}(\tau)}{R_\infty} = \frac{\mathbb{E}\left[\|x_t^\star - \mathbb{E}[x_t^\star \mid \mathcal{G}_t^\tau]\|_Q^2\right]}{\mathbb{E}\left[\|x_t^\star - \mathbb{E}[x_t^\star]\|_Q^2\right]}. \tag{45}$$

Thus

$$R_{\mathrm{glob}}(\tau) = \eta_T(\tau) R_\infty. \tag{46}$$

The function $\eta_T(\tau) \in [0,1]$ measures the fraction of decision-relevant uncertainty that remains when the newest available global state is τ old. It is nondecreasing in τ because the available σ-algebra becomes smaller as the delay increases. At $\tau = 0$,

$$\eta_T(0) = 0, \tag{47}$$

since the current complete state is available and $x_t^\star$ is therefore known exactly. Under an appropriate mixing condition, $\eta_T(\tau) \to 1$ as $\tau \to \infty$; this convergence need not hold for a general stationary process containing perfectly persistent latent information.

Similarly, define the *fresh-local captured share*

$$\eta_{\mathrm{loc}} \triangleq 1 - \frac{R_{\mathrm{loc}}}{R_\infty} \in [0,1]. \tag{48}$$

The quantity η_{loc} measures the fraction of decision-relevant variation that can be captured using fresh-local information, with prediction error weighted by its consequence for the decision objective.

These two quantities give an immediate general crossover characterization.

Theorem 1 (General freshness–locality crossover)**.** *For any stationary environment satisfying the assumptions above,*

$$R_{\mathrm{loc}} < R_{\mathrm{glob}}(\tau) \quad \Longleftrightarrow \quad \eta_T(\tau) > 1 - \eta_{\mathrm{loc}}. \tag{49}$$

Proof. By definitions (46) and (48),

$$R_{\mathrm{glob}}(\tau) = \eta_T(\tau) R_\infty, \qquad R_{\mathrm{loc}} = (1 - \eta_{\mathrm{loc}}) R_\infty.$$

Since $R_\infty > 0$, comparing the two gives (49). □

Although Theorem 1 is algebraically simple, it separates the two fundamentally different sources of architectural loss. The quantity $1-\eta_{\mathrm{loc}}$ measures what is lost by restricting information spatially, while $\eta_T(\tau)$ measures what is lost by allowing global information to age in time. Fresh-local information is preferable precisely when the temporal information lost by waiting for a global view exceeds the decision-relevant information unavailable locally.

3.3 Gauss–Markov Temporal Model

We now specialize the temporal unpredictability function to obtain a closed form. Suppose the environment follows the stationary multivariate Ornstein–Uhlenbeck process

$$d\theta_t = -\frac{1}{T}\left(\theta_t - \bar{\theta}\right) dt + \sqrt{\frac{2}{T}}\, \Sigma_\infty^{1/2} dW_t, \tag{50}$$

where $T > 0$ is the temporal coherence time, $\bar{\theta}$ is the stationary mean, and Σ_∞ is the stationary covariance matrix. The process may exhibit arbitrary instantaneous cross-correlation among agents through Σ_∞; the simplifying assumption is that all temporal modes share the same decay time T.

Assume further that

$$b(\theta) = B\theta + b_0, \tag{51}$$

so that the full-information decision is affine:

$$x^\star(\theta) = Q^{-1}B\theta + Q^{-1}b_0. \tag{52}$$

Let

$$\rho \triangleq \frac{\tau}{T} \tag{53}$$

denote the dimensionless staleness ratio.

The OU Markov property gives

$$\mathbb{E}\left[\theta_t \mid \mathcal{G}_t^\tau\right] = \bar{\theta} + e^{-\rho}\left(\theta_{t-\tau} - \bar{\theta}\right), \tag{54}$$

$$\mathrm{Cov}\left(\theta_t \mid \mathcal{G}_t^\tau\right) = \left(1 - e^{-2\rho}\right)\Sigma_\infty. \tag{55}$$

We can therefore evaluate the stale-global regret exactly.

Theorem 2 (Exact stale-global regret)**.** *Under* (50)–(51),

$$\boxed{R_{\mathrm{glob}}(\tau) = \left(1 - e^{-2\tau/T}\right) R_\infty,} \tag{56}$$

where

$$R_\infty = \frac{1}{2}\,\mathrm{tr}\left(B^\top Q^{-1} B\,\Sigma_\infty\right). \tag{57}$$

Equivalently,

$$\boxed{\eta_T(\tau) = 1 - e^{-2\tau/T}.} \tag{58}$$

Proof. From (52) and (55),

$$\mathrm{Cov}\left(x_t^\star \mid \mathcal{G}_t^\tau\right) = Q^{-1}B\left[\left(1 - e^{-2\tau/T}\right)\Sigma_\infty\right]B^\top Q^{-1}. \tag{59}$$

Substituting into (44) gives

$$R_{\mathrm{glob}}(\tau) = \frac{1}{2}\left(1 - e^{-2\tau/T}\right)\mathrm{tr}\left(B^\top Q^{-1} B \Sigma_\infty\right), \tag{60}$$

which yields (56). The same trace expression without the exponential factor is exactly the open-loop regret (41), giving (57). □

The result has a simple interpretation. When $\tau \ll T$, the environment changes little over the information delay and

$$R_{\mathrm{glob}}(\tau) = 2\frac{\tau}{T}R_\infty + o\left(\frac{\tau}{T}\right). \tag{61}$$

When $\tau \gg T$, the delayed global history becomes essentially uninformative about the current decision and

$$R_{\mathrm{glob}}(\tau) \to R_\infty.$$

Thus the relevant measure of staleness is not delay alone but delay relative to the timescale on which decision-relevant state changes. For any fixed $\tau > 0$, the same identity gives $R_{\mathrm{glob}}(\tau) \to R_\infty$ as $T \downarrow 0$ and $R_{\mathrm{glob}}(\tau) \to 0$ as $T \to \infty$.

3.4 Fresh-Local versus Stale-Global Crossover

Combining Theorem 1 with (58) yields the first main architectural result.

Theorem 3 (Fresh-local/stale-global crossover)**.** *Under the Gauss–Markov and affine-decision assumptions of Theorem 2, fresh-local information strictly outperforms stale-global information if and only if*

$$\boxed{\frac{\tau}{T} > \frac{1}{2}\log\frac{1}{\eta_{\mathrm{loc}}}.} \tag{62}$$

Equivalently, defining

$$\rho^\star \triangleq \frac{1}{2}\log\frac{1}{\eta_{\mathrm{loc}}}, \tag{63}$$

we have

$$R_{\mathrm{loc}} < R_{\mathrm{glob}}(\tau) \quad\iff\quad \rho > \rho^\star. \tag{64}$$

Proof. By Theorem 1, fresh-local wins exactly when

$$1 - e^{-2\rho} > 1 - \eta_{\mathrm{loc}}.$$

Equivalently,

$$e^{-2\rho} < \eta_{\mathrm{loc}},$$

which gives (62). □

The limiting cases are informative. If

$$\eta_{\mathrm{loc}} = 1,$$

fresh-local information is sufficient to reproduce the full-information decision, and local control dominates for every $\tau > 0$. If

$$\eta_{\mathrm{loc}} = 0,$$

local information provides no improvement over open loop, and stale-global information is never strictly worse at any finite delay.

More generally, Theorem 3 identifies two dimensionless determinants of the architecture choice. The staleness ratio τ/T measures the loss of temporal relevance incurred by global information, while η_{loc} measures the share of decision-relevant variation captured locally. Within the comparison of the two pure benchmarks, fresh-local has lower regret precisely when the two loss fractions cross.

3.5 Role of Decision Coupling

The fresh-local captured share η_{loc} depends jointly on the spatial statistics of the environment and on the coupling structure of the decision problem. To isolate the role of decision coupling, consider a symmetric shared-resource model with scalar actions,

$$f(x,\theta) = \sum_{i=1}^{n}\left(\frac{q}{2}x_i^2 - \theta_i x_i\right) + \frac{\gamma}{2}\left(\sum_{i=1}^{n} x_i\right)^2, \qquad q > 0,\ \gamma \geq 0. \tag{65}$$

Here q determines the local curvature or stiffness of each decision, while γ determines the strength of the shared coordination penalty. In the notation of Section 2,

$$Q = qI + \gamma \mathbf{1}\mathbf{1}^\top, \qquad b(\theta) = \theta. \tag{66}$$

Assume that the state components θ_i are independent, mean-zero stationary OU processes with common variance σ^2 and common coherence time T.

The full-information action is

$$x_i^\star = \frac{\theta_i}{q} - \frac{\gamma}{q(q+\gamma n)} \sum_{j=1}^{n} \theta_j. \tag{67}$$

Thus, even though the linear reward $\theta_i x_i$ is local, coupling through the common resource makes agent i's optimal action depend on the states of all other agents.

Proposition 2 (Local explainability under aggregate coupling)**.** *For the model* (65)*, the optimal fresh-local policy is*

$$u_i^{\mathrm{loc}} = \frac{\theta_i}{q+\gamma}, \tag{68}$$

and

$$R_\infty = \frac{n\sigma^2\,(q+\gamma(n-1))}{2q(q+\gamma n)}, \tag{69}$$

$$R_{\mathrm{loc}} = \frac{n\sigma^2\gamma^2(n-1)}{2q(q+\gamma)(q+\gamma n)}. \tag{70}$$

Consequently, the fresh-local captured share is

$$\boxed{\eta_{\mathrm{loc}} = \frac{q(q+\gamma n)}{(q+\gamma)\,(q+\gamma(n-1))}.} \tag{71}$$

Proof. Let $\mu_j = \mathbb{E}[u_j]$. Because the component processes are independent, for $j \neq i$ the full local history satisfies

$$\mathbb{E}[u_j \mid \mathcal{G}_{i,t}^{\mathrm{loc}}] = \mathbb{E}[u_j] = \mu_j,$$

while $\theta_{i,t}$ is $\mathcal{G}_{i,t}^{\mathrm{loc}}$-measurable. The conditional normal equation (35) therefore gives

$$(q+\gamma)u_i + \gamma \sum_{j\neq i} \mu_j - \theta_{i,t} = 0.$$

Taking expectations yields

$$q\mu_i + \gamma \sum_j \mu_j = 0.$$

Summing over i implies $\sum_j \mu_j = 0$, hence every $\mu_i = 0$ and $u_i = \theta_{i,t}/(q+\gamma)$, which proves (68). This argument uses independence, zero means, square integrability, and strict convexity; Gaussianity is not required for the local-policy formula.

The expressions (69) and (70) follow by substituting the full-information and local policies into the quadratic regret expression (32). Substituting them into (48) and simplifying yields (71). □

Combining Proposition 2 with Theorem 3 gives an explicit threshold:

$$\boxed{\rho^\star(\gamma, n) = \frac{1}{2}\log\left[\frac{(q+\gamma)\,(q+\gamma(n-1))}{q(q+\gamma n)}\right].} \tag{72}$$

Therefore

$$R_{\mathrm{loc}} < R_{\mathrm{glob}}(\tau) \iff \frac{\tau}{T} > \rho^\star(\gamma, n). \tag{73}$$

As $n \to \infty$,

$$\boxed{\rho^\star(\gamma, n) \longrightarrow \frac{1}{2}\log\left(1+\frac{\gamma}{q}\right).} \tag{74}$$

The local–global architecture choice therefore collapses, in the large-system limit, to a comparison between two dimensionless quantities:

$$\boxed{\underbrace{\frac{\tau}{T}}_{\text{information staleness}} \qquad \text{and} \qquad \underbrace{\frac{1}{2}\log\left(1+\frac{\gamma}{q}\right)}_{\text{value of coordination}}.} \tag{75}$$

When $\gamma = 0$, the decisions decouple and $\eta_{\text{loc}} = 1$: current local information is sufficient, so any positive global delay makes fresh-local control preferable. As γ increases, each agent's optimal action depends more strongly on remote states, η_{loc} decreases, and the crossover moves to larger τ/T. Thus stronger coordination coupling makes the system willing to tolerate increasingly stale-global information before abandoning global coordination in favor of fresh-local decisions.

The result provides the first endpoint characterization of the broader freshness–scope problem. The next sections move beyond the binary local–global comparison by allowing the information radius itself to vary, and ask how spatial predictability and temporal staleness jointly determine the optimal coordination scale.

4 Spatial and Temporal Predictability

The preceding section compared the two endpoints of the freshness–scope tradeoff: fresh-local information and stale-global information. We now introduce a family of intermediate architectures in which each agent may use information from within a spatial radius r. The purpose of this section is to characterize the two forms of predictability that determine the value of such an architecture.

Temporal predictability determines how much decision-relevant information is lost while observations are being communicated. Spatial predictability determines how much of the globally optimal decision can be inferred without observing the entire network. The relevant quantities are not raw state prediction errors alone, but prediction errors weighted by their effect on the full-information decision.

4.1 Spatial Structure of the Environment

For simplicity of exposition, first suppose that each node has a scalar local state, so that

$$\theta_t = (\theta_{1,t}, \ldots, \theta_{n,t})^\top.$$

The vector-valued case follows by replacing scalar covariances with block covariances.

Let

$$\bar{\theta} \triangleq \mathbb{E}[\theta_t], \qquad \Sigma_S \triangleq \operatorname{Cov}(\theta_t) \succeq 0 \tag{76}$$

denote the stationary mean and spatial covariance. The entry $(\Sigma_S)_{ij}$ measures the instantaneous statistical dependence between the states associated with nodes i and j.

On a path, line, or Euclidean spatial domain, a canonical covariance model is

$$\operatorname{Cov}(\theta_{i,t}, \theta_{j,t}) = \sigma^2 \exp\left(-\frac{d_G(i,j)}{\ell_s}\right), \tag{77}$$

where $\ell_s > 0$ is a spatial correlation length. Larger ℓ_s means that states remain strongly correlated over greater network distance. In one spatial dimension, (77) is the exponential, or Ornstein–Uhlenbeck, covariance model [19].

For an arbitrary graph, a convenient alternative is to define the covariance spectrally through the graph Laplacian L_G. For example,

$$\Sigma_S = \sigma^2 \left(\kappa_s^2 I + L_G\right)^{-\nu}, \qquad \kappa_s > 0, \quad \nu > 0, \tag{78}$$

is positive semidefinite and places more state energy in low graph-frequency modes. Such constructions are closely related to graph signal smoothness and Gaussian Markov random-field models [20, 21].

A basic graph-smoothness statistic is

$$\begin{aligned} \mathcal{E}_G &\triangleq \mathbb{E}\left[(\theta_t - \bar{\theta})^\top L_G(\theta_t - \bar{\theta})\right] \\ &= \operatorname{tr}\left(L_G \Sigma_S\right). \end{aligned} \tag{79}$$

For an undirected weighted graph,

$$(\theta_t - \bar{\theta})^\top L_G(\theta_t - \bar{\theta}) = \frac{1}{2}\sum_{i,j} w_{ij}\left[(\theta_{i,t} - \bar{\theta}_i) - (\theta_{j,t} - \bar{\theta}_j)\right]^2. \tag{80}$$

Small $\mathcal{E}_G$ therefore indicates that nearby nodes tend to have similar states.

Neither ℓ_s nor $\mathcal{E}_G$ by itself determines architecture regret. A state component may be difficult to infer spatially yet have little effect on the desired decision. We therefore introduce a decision-relevant spatial predictability function below.

4.2 Temporal Structure

Section 3 defined the normalized temporal unpredictability function

$$\eta_T(\tau) = \frac{R_{\text{glob}}(\tau)}{R_\infty}. \tag{81}$$

It measures the fraction of full-information decision variation that cannot be predicted from global information whose age is τ.

The general framework does not require a particular temporal process. The function $\eta_T(\tau)$ may be obtained empirically or calculated from any specified stochastic model. For the exact results in this and the following section, however, we use the common-rate Gauss–Markov model

$$d\theta_t = -\frac{1}{T}\left(\theta_t - \bar{\theta}\right)dt + \sqrt{\frac{2}{T}}\,\Sigma_S^{1/2} dW_t. \tag{82}$$

Its stationary space-time covariance is

$$\operatorname{Cov}\left(\theta_t, \theta_s\right) = e^{-|t-s|/T}\Sigma_S. \tag{83}$$

Thus Σ_S specifies arbitrary instantaneous spatial dependence, while T specifies a common temporal coherence time.

Equation (83) is a separable space-time model. Separability is used here because it produces an exact composition law between spatial omission and temporal staleness. More general nonseparable covariance models can be accommodated by working directly with a joint prediction-error function, although the factorization below will not generally remain exact; see, for example, [22].

4.3 Joint Spatio-Temporal Information

For radius r, define the zero-delay information available to agent i at time t as

$$\mathcal{I}_{i,t}^{(r)} \triangleq \sigma(\theta_{j,t} : j \in \mathcal{N}_r(i)) . \tag{84}$$

The corresponding zero-delay radius-r architecture is

$$\mathcal{A}_{r,0} \triangleq \left(\mathcal{I}_{1,t}^{(r)}, \ldots, \mathcal{I}_{n,t}^{(r)}\right) . \tag{85}$$

A delayed synchronized-snapshot radius-r architecture provides the same type of information, shifted backward by τ:

$$\mathcal{A}_{r,\tau} \triangleq \left(\mathcal{I}_{1,t-\tau}^{(r)}, \ldots, \mathcal{I}_{n,t-\tau}^{(r)}\right) . \tag{86}$$

For comparison, define the distinct history information set

$$\widetilde{\mathcal{I}}_{i,t}^{(r)} \triangleq \sigma(\theta_{j,s} : j \in \mathcal{N}_r(i),\ s \le t) . \tag{87}$$

The principal architecture-design claims use the synchronized-snapshot family (86). Theorem 4 also applies to the consistently shifted history family, with its own zero-delay omission function.

Let

$$\mathcal{F}_t \triangleq \sigma(\theta_s : s \le t) \tag{88}$$

and define the centered admissible-policy subspace

$$\mathcal{S}_0(\mathcal{A}) \triangleq \{u \in \mathcal{S}(\mathcal{A}) : \mathbb{E}[u] = 0\}. \tag{89}$$

A delayed architecture family is *shift-consistent* if time translation by τ maps the zero-delay centered admissible-policy subspace isometrically onto the corresponding delayed subspace. We write $P_{r,\tau}$ for the Q-orthogonal projection onto $\mathcal{S}_0(\mathcal{A}_{r,\tau})$.

Lemma 2 (Snapshot–history sufficiency under a common temporal rate)**.** *Suppose the process is Gaussian and*

$$\mathrm{Cov}(\theta_t, \theta_s) = e^{-|t-s|/T}\Sigma_S. \tag{90}$$

For an observed component set N with complement $\bar{N}$, $s < t$, and nonsingular Σ_{NN},

$$\begin{aligned} &\mathrm{Cov}\big(\theta_{\bar{N},t}, \theta_{N,s} \mid \theta_{N,t}\big) \\ &\quad = e^{-(t-s)/T}\Sigma_{\bar{N}N} - \Sigma_{\bar{N}N}\Sigma_{NN}^{-1}e^{-(t-s)/T}\Sigma_{NN} = 0. \end{aligned} \tag{91}$$

Consequently,

$$\theta_{\bar{N},t} \perp\!\!\!\perp \sigma(\theta_{N,s} : s < t) \mid \theta_{N,t}. \tag{92}$$

For affine current-state decisions, the current observed snapshot is therefore sufficient relative to the observed history for predicting unobserved current components, and the two variants have the same zero-delay decision-relevant prediction error. If Σ_{NN} is singular, the same statement uses the Moore–Penrose inverse on the covariance support. The result need not hold for heterogeneous temporal rates or nonseparable space–time processes.

Proof. The displayed conditional covariance follows from the Gaussian conditional covariance formula and the common scalar decay factor. Gaussianity converts zero conditional cross-covariance into conditional independence; affine prediction then gives the stated decision-level sufficiency. □

Define the normalized zero-delay spatial omission function

$$\eta_S(r) \triangleq \frac{R(\mathcal{A}_{r,0})}{R_\infty}. \tag{93}$$

The quantity $\eta_S(r)$ is the fraction of decision-relevant variation that cannot be captured using current information within radius r.

Since zero-delay radius architectures are nested,

$$r_2 \geq r_1 \quad \Longrightarrow \quad \mathcal{S}(\mathcal{A}_{r_1,0}) \subseteq \mathcal{S}(\mathcal{A}_{r_2,0}), \tag{94}$$

and therefore

$$0 \leq \eta_S(r_2) \leq \eta_S(r_1) \leq 1. \tag{95}$$

If radius zero observes precisely the local current component, then

$$\eta_S(0) = \frac{R_{\text{loc}}}{R_\infty} = 1 - \eta_{\text{loc}}. \tag{96}$$

Under the common-rate Gaussian-affine assumptions, Lemma 2 identifies this snapshot endpoint with the history-based fresh-local benchmark for the current prediction problem. No such identification is asserted for a general stationary process. On a finite graph with diameter D, complete current state is available at $r = D$, so

$$\eta_S(D) = 0. \tag{97}$$

4.4 Decision-Relevant Smoothness

Suppose, as in (15), that

$$b(\theta) = B\theta + b_0. \tag{98}$$

Then

$$x_t^\star = K\theta_t + k_0, \qquad K \triangleq Q^{-1}B, \qquad k_0 \triangleq Q^{-1}b_0. \tag{99}$$

The matrix K is the decision-sensitivity operator: its block K_{ij} describes how the state at node j influences the full-information decision at node i.

The covariance of the clairvoyant decision is

$$\Sigma_x \triangleq \mathrm{Cov}(x_t^\star) = K\Sigma_S K^\top. \tag{100}$$

Thus spatial state modes that lie approximately in the null space of K have little decision value, even if they are statistically unpredictable. By contrast, uncertainty in a state mode strongly amplified by K can dominate architecture regret.

The function $\eta_S(r)$ in (93) incorporates all three relevant ingredients:

$$\boxed{\text{state spatial statistics} \quad + \quad \text{decision sensitivity} \quad + \quad \text{information geometry.}} \tag{101}$$

It is therefore more informative for architecture design than a raw correlation length alone. The next result shows that, under the common-rate Gauss–Markov model, $\eta_S(r)$ and temporal unpredictability combine in an especially simple way.

Theorem 4 (Spatio-temporal predictability composition law). *Suppose the environment follows* (82)*, the full-information decision is affine as in* (99)*, and the radius-dependent information family is shift-consistent. Then, for every radius r and delay τ,*

$$\boxed{R(\mathcal{A}_{r,\tau}) = \left(1 - e^{-2\tau/T}\right) R_\infty + e^{-2\tau/T} R(\mathcal{A}_{r,0}).} \tag{102}$$

Equivalently,

$$\boxed{\frac{R(\mathcal{A}_{r,\tau})}{R_\infty} = 1 - e^{-2\tau/T} \left(1 - \eta_S(r)\right).} \tag{103}$$

In terms of the temporal unpredictability $\eta_T(\tau) = 1 - e^{-2\tau/T}$,

$$\boxed{\eta_{ST}(r,\tau) = \eta_T(\tau) + \left(1 - \eta_T(\tau)\right) \eta_S(r),} \tag{104}$$

where

$$\eta_{ST}(r,\tau) \triangleq \frac{R(\mathcal{A}_{r,\tau})}{R_\infty}.$$

Proof. Let

$$\bar{x}^\star \triangleq \mathbb{E}[x_t^\star], \qquad X_t \triangleq x_t^\star - \bar{x}^\star. \tag{105}$$

Because constant policies are feasible under every architecture, architecture regret depends only on the centered process X_t.

Under (82) and (99),

$$X_t = e^{-\tau/T} X_{t-\tau} + \xi_{t,\tau}, \tag{106}$$

where

$$\mathbb{E}\left[\xi_{t,\tau} \mid \mathcal{F}_{t-\tau}\right] = 0 \tag{107}$$

and

$$\mathbb{E}\left[\|\xi_{t,\tau}\|_Q^2\right] = \left(1 - e^{-2\tau/T}\right) \mathbb{E}\left[\|X_t\|_Q^2\right]. \tag{108}$$

Let $P_{r,\tau}$ be the Q-orthogonal projection defined above onto the centered policy subspace induced by $\mathcal{A}_{r,\tau}$. Every policy in this subspace is measurable with respect to $\mathcal{F}_{t-\tau}$. Hence (107) implies that $\xi_{t,\tau}$ is orthogonal to the entire admissible subspace. By linearity of orthogonal projection,

$$P_{r,\tau} X_t = e^{-\tau/T} P_{r,\tau} X_{t-\tau}. \tag{109}$$

Consequently,

$$X_t - P_{r,\tau} X_t = e^{-\tau/T} \left(X_{t-\tau} - P_{r,\tau} X_{t-\tau}\right) + \xi_{t,\tau}. \tag{110}$$

The two terms on the right-hand side are orthogonal, so

$$\begin{aligned} \mathbb{E}\left[\|X_t - P_{r,\tau} X_t\|_Q^2\right] = e^{-2\tau/T} \mathbb{E}\left[\|X_{t-\tau} - P_{r,\tau} X_{t-\tau}\|_Q^2\right] \\ + \mathbb{E}\left[\|\xi_{t,\tau}\|_Q^2\right]. \end{aligned} \tag{111}$$

By stationarity and shift consistency,

$$\frac{1}{2} \mathbb{E}\left[\|X_{t-\tau} - P_{r,\tau} X_{t-\tau}\|_Q^2\right] = R(\mathcal{A}_{r,0}), \tag{112}$$

Indeed, time translation maps the zero-delay centered policy subspace isometrically onto the corresponding subspace at $t-\tau$, so the projection residual has the same distribution and weighted norm. Moreover,

$$\frac{1}{2}\mathbb{E}\left[\|X_t\|_Q^2\right] = R_\infty. \tag{113}$$

Substitution of (108) into (111) gives (102); normalization by R_∞ gives (103) and (104). $\square$

Theorem 4 gives a precise meaning to the two architectural errors. The term

$$\left(1 - e^{-2\tau/T}\right) R_\infty$$

is the temporal innovation that no information available at time $t-\tau$ can predict. The remaining fraction $e^{-2\tau/T}$ of the current decision is temporally predictable, but only the fraction $1-\eta_S(r)$ of that component can be reconstructed from radius-r information.

Spatial omission and temporal staleness are therefore not simply added as two independent penalties. Spatial information can only help with the portion of the current decision that remains temporally predictable when it arrives.

5 Optimal Coordination Scale

We now make information radius an architectural design variable. For each r, an agent may base its action on observations available within radius r, but those observations arrive after latency $\tau(r)$. The objective is to select the spatial scope that minimizes architecture regret.

On a graph, admissible radii are generally integers. We first analyze the continuous relaxation $r \in [0, D]$, where D is the maximum allowed information radius. On a finite graph, the exact optimizer is obtained by minimizing over the admissible integer radii. A neighboring-integer rule is justified only when a specified continuous interpolation is unimodal.

5.1 Radius-Dependent Information Architectures

Let

$$\tau : [0, D] \to [0, \infty) \tag{114}$$

be a nondecreasing information-propagation law satisfying

$$\tau(0) = 0. \tag{115}$$

Architecture $\mathcal{A}_{r,\tau(r)}$ provides each agent with information from a synchronized radius-r snapshot delayed by $\tau(r)$.

The synchronized-snapshot family is not nested in radius: increasing r replaces a narrower, fresher snapshot with a broader, older one. This replacement, rather than the mere addition of remote observations, is what permits an interior optimum in the absence of an explicit information-acquisition cost. Under a native-age or streaming architecture that retained all previously available observations and only added remote observations as r increased, the information sets would be nested and architecture regret would be weakly nonincreasing in r.

Define

$$R(r) \triangleq R\left(\mathcal{A}_{r,\tau(r)}\right). \tag{116}$$

By Theorem 4,

$$\boxed{R(r) = R_\infty \left[1 - e^{-2\tau(r)/T}\left(1 - \eta_S(r)\right)\right].} \tag{117}$$

On a finite graph, moving from radius r to $r+1$ strictly improves performance exactly when

$$e^{-2\tau(r+1)/T}\big(1-\eta_S(r+1)\big) > e^{-2\tau(r)/T}\big(1-\eta_S(r)\big), \tag{118}$$

or equivalently

$$\log\left(\frac{1-\eta_S(r+1)}{1-\eta_S(r)}\right) > \frac{2\big(\tau(r+1)-\tau(r)\big)}{T}. \tag{119}$$

Increasing the radius by one hop is beneficial precisely when the discrete proportional gain in decision-relevant spatial information exceeds the freshness loss incurred over that hop.

At radius zero,

$$R(0) = R_\infty \eta_S(0) = R_{\text{loc}}. \tag{120}$$

On a finite graph, if $D = \operatorname{diam}(G)$ and radius D provides the entire state, then

$$R(D) = R_\infty \left(1 - e^{-2\tau(D)/T}\right) = R_{\text{glob}}(\tau(D)). \tag{121}$$

Under the common-rate Gaussian-affine model, Lemma 2 makes the snapshot endpoints in (120)–(121) equivalent for the current prediction problem to the history-based pure benchmarks of Section 3. They are not literally the same information sets for a general stationary process.

5.2 Spatial-Omission versus Temporal-Staleness Tradeoff

Equation (117) can be written as

$$R(r) = R_T(r) + R_S(r), \tag{122}$$

where

$$R_T(r) \triangleq \left(1 - e^{-2\tau(r)/T}\right) R_\infty, \tag{123}$$

$$R_S(r) \triangleq e^{-2\tau(r)/T}\eta_S(r) R_\infty. \tag{124}$$

The temporal term increases with information radius whenever $\tau(r)$ does. The zero-delay spatial omission fraction $\eta_S(r)$ decreases with radius, although its contribution is discounted by the temporal relevance factor $e^{-2\tau(r)/T}$.

To characterize the optimum, define the marginal temporal decay rate

$$m_T(r) \triangleq \frac{2\tau'(r)}{T} \tag{125}$$

and the proportional marginal spatial information gain

$$m_S(r) \triangleq -\frac{\eta_S'(r)}{1-\eta_S(r)}. \tag{126}$$

The latter quantity is the rate at which the spatially explained share $1-\eta_S(r)$ grows:

$$m_S(r) = \frac{d}{dr}\log\left(1-\eta_S(r)\right). \tag{127}$$

Theorem 5 (Continuous marginal-balance characterization). *Under the assumptions of Theorem 4, additionally assume that $\tau(r)$ and $\eta_S(r)$ are continuously differentiable on $[0, D]$, with*

$$\tau'(r) \geq 0, \qquad \eta_S'(r) \leq 0, \qquad 0 \leq \eta_S(r) < 1.$$

Then

$$R'(r) = R_\infty e^{-2\tau(r)/T} \left(1 - \eta_S(r)\right) \left[m_T(r) - m_S(r)\right]. \tag{128}$$

Consequently, every interior optimal radius $r^\star \in (0, D)$ satisfies

$$\boxed{-\frac{\eta_S'(r^\star)}{1 - \eta_S(r^\star)} = \frac{2\tau'(r^\star)}{T}.} \tag{129}$$

Suppose additionally that $m_S(r)$ is strictly decreasing and $m_T(r)$ is nondecreasing. Then $R(r)$ is unimodal and exactly one of the following holds:

1. *If*
$$m_S(0) \leq m_T(0), \tag{130}$$
then $r^\star = 0$.

2. *If*
$$m_S(D) \geq m_T(D), \tag{131}$$
then $r^\star = D$.

3. *Otherwise, there is a unique $r^\star \in (0, D)$ satisfying* (129)*.*

Proof. Differentiating (117) gives

$$\begin{aligned} \frac{R'(r)}{R_\infty} &= e^{-2\tau(r)/T} \left[\frac{2\tau'(r)}{T} \left(1 - \eta_S(r)\right) + \eta_S'(r)\right] \\ &= e^{-2\tau(r)/T} \left(1 - \eta_S(r)\right) \left[m_T(r) - m_S(r)\right], \end{aligned} \tag{132}$$

which proves (128) and the interior first-order condition.

Under the additional assumptions, the difference

$$m_T(r) - m_S(r)$$

is strictly increasing. It can therefore change sign at most once. If it is nonnegative at $r = 0$, regret is nondecreasing and radius zero is optimal. If it is nonpositive at $r = D$, regret is nonincreasing and radius D is optimal. Otherwise it crosses zero exactly once, from negative to positive, yielding a unique interior minimizer. $\square$

Within the common-rate synchronized-snapshot model, Theorem 5 gives the marginal-balance characterization

$$\boxed{\text{expand the information radius until} \quad \text{marginal spatial value} = \text{marginal freshness loss.}} \tag{133}$$

The first-order condition applies to any differentiable latency law, but uniqueness requires the stated single-crossing assumptions. Concave aggregation laws can make $m_T(r)$ decrease with radius, in which case multiple crossings or boundary optima cannot be excluded; convex congestion laws act in the opposite direction.

It also gives comparative statics without committing to a specific covariance model. Under the single-crossing conditions of the theorem:

- Increasing T lowers $m_T(r)$ pointwise and therefore moves the optimum toward a larger radius.
- Increasing communication latency, for example by replacing $\tau(r)$ with $a\tau(r)$ for $a > 1$, raises $m_T(r)$ and moves the optimum toward a smaller radius.
- Any increase in decision coupling that raises $m_S(r)$ pointwise makes additional spatial information more valuable and moves the optimum toward a larger radius.
- Any increase in spatial predictability that lowers $m_S(r)$ pointwise makes additional radius less valuable and moves the optimum toward a smaller radius.

The final two statements are single-crossing conditions on the decision-relevant function $\eta_S(r)$; they are not automatic consequences of changing an arbitrary raw covariance parameter.

5.3 An Exponential Spatial-Predictability Regime

A simple but useful closed form is obtained when the zero-delay spatial omission fraction decays exponentially:

$$\eta_S(r) = \eta_0 e^{-2r/\ell_c}, \qquad 0 < \eta_0 < 1. \tag{134}$$

Here η_0 is the decision variation not explained by radius-zero information, and ℓ_c is a decision-relevance length: larger ℓ_c means that decision-relevant information remains distributed over a greater distance.

Assume information propagates at effective speed $v > 0$, so that

$$\tau(r) = \frac{r}{v}. \tag{135}$$

The distance

$$L_T \triangleq vT \tag{136}$$

is the distance over which information can propagate during one temporal coherence time.

Proposition 3 (Optimal radius under exponential spatial predictability)**.** *Under* (134) *and* (135)*, the unique optimal radius on* $[0, \infty)$ *is*

$$\boxed{r^\star = \frac{\ell_c}{2}\left[\log\left(\eta_0\left[1 + \frac{L_T}{\ell_c}\right]\right)\right]_+}, \tag{137}$$

where $[z]_+ \triangleq \max\{z, 0\}$*. If the architecture imposes a maximum radius* D*, the optimizer is*

$$r_D^\star = \min\{D, r^\star\}. \tag{138}$$

Proof. Under (134),

$$m_S(r) = \frac{2\eta_0 e^{-2r/\ell_c}}{\ell_c\left(1 - \eta_0 e^{-2r/\ell_c}\right)}, \tag{139}$$

while

$$m_T(r) = \frac{2}{L_T}. \tag{140}$$

The marginal-balance equation gives

$$\eta_0 e^{-2r^\star/\ell_c} = \frac{\ell_c}{\ell_c + L_T}. \tag{141}$$

Solving for $r^\star$ yields (137). If the logarithm is nonpositive, $R'(0) \geq 0$ and radius zero is optimal. Unimodality follows from Theorem 5. □

The dimensionless form is

$$\boxed{\frac{r^\star}{\ell_c} = \frac{1}{2}\left[\log\left(\eta_0\left[1+\frac{L_T}{\ell_c}\right]\right)\right]_+.} \tag{142}$$

Thus optimal scope depends on the ratio between the temporal propagation length L_T and the decision-relevance length ℓ_c, together with the fraction η_0 of decision variation unavailable locally.

5.4 A Canonical Spatio-Temporal Field Model

We now derive the exponential form (134) from a concrete stochastic field and decision model. Consider a dense one-dimensional network, represented by locations $z \in \mathbb{R}$. This is also the continuum approximation of a sufficiently long path graph away from its boundaries.

Let $\theta(z,t)$ be a zero-mean Gaussian field with covariance

$$\mathbb{E}\left[\theta(z,t)\theta(z',s)\right] = \sigma^2 \exp\left(-\frac{|z-z'|}{\ell_s}\right)\exp\left(-\frac{|t-s|}{T}\right). \tag{143}$$

The parameter ℓ_s is the spatial correlation length, and T is the temporal coherence time.

Suppose the full-information decision at location z is the exponentially weighted spatial aggregate

$$x^\star(z,t) = \int_{-\infty}^{\infty} w_{\ell_c}(u)\theta(z+u,t)\,du, \tag{144}$$

where

$$w_{\ell_c}(u) \triangleq \frac{1}{2\ell_c}\exp\left(-\frac{|u|}{\ell_c}\right). \tag{145}$$

The decision-relevance length ℓ_c controls how broadly remote states influence the desired local action.

For example, (144) is generated pointwise by the quadratic tracking-loss density

$$f_z(x,\theta) = \frac{q}{2}\left[x(z) - x^\star(z,\theta)\right]^2, \qquad q>0. \tag{146}$$

All regrets in this translation-invariant infinite-line model are understood per unit length, equivalently as the limit of spatially averaged regret on expanding finite intervals. This convention avoids assigning an infinite total cost to a stationary field on $\mathbb{R}$. Equivalently, in the notation of Section 2, $Q = qI$ and $b(\theta) = qK_{\ell_c}\theta$, where K_{ℓ_c} is convolution with w_{ℓ_c}. Thus the canonical model places nonlocal decision dependence in $b(\theta)$. Direct action coupling through a non-diagonal Q changes the specific form of $\eta_S(r)$ but not the general composition law or Theorem 5.

At zero delay, the radius-r observation at location z is

$$\mathcal{O}_r(z,t) \triangleq \sigma\left(\theta(z+u,t) : |u| \le r\right). \tag{147}$$

Proposition 4 (Spatial omission in the canonical line model)**.** *For the field* (143)*, decision* (144)*, and observation* (147)*, the normalized zero-delay spatial omission error is*

$$\boxed{\eta_S(r) = \frac{\ell_c}{\ell_c + 2\ell_s}\exp\left(-\frac{2r}{\ell_c}\right).} \tag{148}$$

In particular,

$$\eta_0 = \eta_S(0) = \frac{\ell_c}{\ell_c+2\ell_s}. \tag{149}$$

Proof. By spatial stationarity, take $z = 0$. Let

$$\kappa \triangleq \frac{1}{\ell_c}, \qquad \lambda \triangleq \frac{1}{\ell_s}. \tag{150}$$

At a fixed time, $\theta(\cdot, t)$ is a stationary one-dimensional Ornstein–Uhlenbeck field and hence is Markov in the spatial coordinate. Conditioned on the field over $[-r, r]$, uncertainty in the two exterior regions is carried by independent left and right spatial innovations.

For the right exterior region, the conditional residual covariance at distances $u, v \geq 0$ beyond the boundary r is

$$\sigma^2 \left[e^{-\lambda|u-v|} - e^{-\lambda(u+v)} \right]. \tag{151}$$

The corresponding contribution to the conditional variance of $x^\star(0, t)$ is

$$\begin{aligned} V_R(r) &= \frac{\kappa^2\sigma^2}{4} e^{-2\kappa r} \int_0^\infty \int_0^\infty e^{-\kappa(u+v)} \\ &\qquad \times \left[e^{-\lambda|u-v|} - e^{-\lambda(u+v)} \right] du\, dv \\ &= \frac{\sigma^2 \kappa \lambda}{4(\kappa+\lambda)^2} e^{-2\kappa r}. \end{aligned} \tag{152}$$

The left exterior region contributes the same amount, so

$$\mathrm{Var}\left(x^\star(0,t) \mid \mathcal{O}_r(0,t)\right) = \frac{\sigma^2 \kappa \lambda}{2(\kappa+\lambda)^2} e^{-2\kappa r}. \tag{153}$$

A direct covariance calculation gives the unconditional variance

$$\mathrm{Var}\left(x^\star(0,t)\right) = \frac{\sigma^2 \kappa \left(2\kappa + \lambda\right)}{2(\kappa+\lambda)^2}. \tag{154}$$

Since the loss in (146) is squared error, the normalized spatial regret is the ratio of (153) to (154). Therefore

$$\eta_S(r) = \frac{\lambda}{2\kappa + \lambda} e^{-2\kappa r} = \frac{\ell_c}{\ell_c + 2\ell_s} e^{-2r/\ell_c}, \tag{155}$$

which proves the claim. □

Remark 2 (Continuum-field interpretation)**.** *The canonical line model can be formulated directly in the L^2 Hilbert space of stationary random fields with the per-unit-length inner product, or as the bulk limit of finite periodic lattices whose circumference tends to infinity. Because the canonical loss has $Q = qI$, the radius-constrained pointwise optimizer is the conditional expectation of $x^\star(z,t)$ given the radius-r field observation, and per-unit-length regret is the corresponding conditional variance. Proposition 4 follows directly in the field formulation or as this finite-lattice limit.*

The expression has the expected limiting behavior. If $\ell_c \to 0$, the desired action becomes purely local and $\eta_S(r) \to 0$ for every $r \geq 0$. If $\ell_s \to \infty$, the state becomes nearly constant over space and a local observation is sufficient to predict the remote field, again driving $\eta_S(r)$ to zero. Conversely, spatially rough states and long-range decision dependence increase the value of nonlocal information.

5.5 Main Result: Closed-Form Optimal Coordination Radius

Combining Proposition 3 and Proposition 4 yields the second main result.

Theorem 6 (Optimal synchronized-snapshot radius in the canonical model)**.** *Suppose the state field obeys* (143)*, the full-information decision is given by* (144)*, and information from radius r arrives after*

$$\tau(r) = \frac{r}{v}, \qquad v > 0. \tag{156}$$

Let

$$L_T = vT \tag{157}$$

be the temporal propagation length. Then the optimal information radius on the infinite line is

$$\boxed{r^\star = \frac{\ell_c}{2}\left[\log\left(\frac{\ell_c + L_T}{\ell_c + 2\ell_s}\right)\right]_+.} \tag{158}$$

Equivalently,

$$\boxed{\frac{r^\star}{\ell_c} = \frac{1}{2}\left[\log\left(\frac{1 + L_T/\ell_c}{1 + 2\ell_s/\ell_c}\right)\right]_+.} \tag{159}$$

The optimal radius is strictly positive if and only if

$$\boxed{L_T > 2\ell_s.} \tag{160}$$

Within the interior regime, $r^\star$:

1. *increases with the temporal coherence time T;*
2. *increases with the information-propagation speed v;*
3. *decreases with the spatial correlation length ℓ_s; and*
4. *increases with the decision-relevance length ℓ_c.*

Proof. Substituting

$$\eta_0 = \frac{\ell_c}{\ell_c + 2\ell_s}$$

and $L_T = vT$ into (137) gives

$$\begin{aligned} r^\star &= \frac{\ell_c}{2}\left[\log\left(\frac{\ell_c}{\ell_c + 2\ell_s}\left[1 + \frac{L_T}{\ell_c}\right]\right)\right]_+ \\ &= \frac{\ell_c}{2}\left[\log\left(\frac{\ell_c + L_T}{\ell_c + 2\ell_s}\right)\right]_+. \end{aligned} \tag{161}$$

The logarithm is positive exactly when $L_T > 2\ell_s$, proving (160).

In the interior regime,

$$\frac{\partial r^\star}{\partial L_T} = \frac{\ell_c}{2(\ell_c + L_T)} > 0, \tag{162}$$

and

$$\frac{\partial r^\star}{\partial \ell_s} = -\frac{\ell_c}{\ell_c + 2\ell_s} < 0. \tag{163}$$

Since $L_T = vT$, the first derivative establishes monotonicity in both v and T.

Finally, when $L_T > 2\ell_s$,

$$r^\star = \frac{1}{2}\int_{2\ell_s}^{L_T} \frac{\ell_c}{\ell_c + s} ds. \tag{164}$$

The integrand is strictly increasing in ℓ_c, proving that $r^\star$ increases with the range over which remote state affects the desired decision. □

Theorem 6 identifies three physical length scales:

$$\boxed{\underbrace{\ell_s}_{\text{spatial correlation}} \qquad \underbrace{\ell_c}_{\text{decision relevance}} \qquad \underbrace{L_T = vT}_{\text{temporal propagation}}}. \tag{165}$$

The temporal propagation length L_T—equivalently, a freshness horizon measured in distance—is how far information can travel before the environment substantially changes. The spatial correlation length ℓ_s measures how much of the remote field can already be predicted locally. The decision-relevance length ℓ_c measures how far remote state continues to affect the desired action.

The threshold $L_T > 2\ell_s$ has a direct interpretation in the canonical model. If information cannot propagate beyond approximately two spatial correlation lengths within one coherence time, then the additional state it would reveal is not worth the staleness incurred in obtaining it, and fresh local information is optimal. Once the temporal propagation length exceeds that scale, a positive information radius becomes beneficial. The preferred radius then grows with the excess propagation horizon and with the spatial range of the decision dependence.

The remaining limiting cases are consistent with this interpretation. With all other parameters fixed, $T \downarrow 0$ or $v \downarrow 0$ gives $L_T \downarrow 0$ and hence $r^\star = 0$; increasing either T or v enlarges the interior optimum, which grows without bound on the infinite line as $L_T \to \infty$. Also, $r^\star \to 0$ as $\ell_c \downarrow 0$, because the desired decision becomes local, and $r^\star = 0$ for sufficiently large ℓ_s (in particular as $\ell_s \to \infty$), because remote state is then predictable locally. On a finite graph these limits are capped by the diameter.

On a finite graph, Theorem 4 and direct minimization over the admissible integer radii remain exact. Theorem 5 is a continuous relaxation that supplies comparative-static and marginal-balance intuition. Its exact finite-graph analogue is the one-hop comparison (118)–(119). Equation (158) is the closed-form homogeneous-line scaling law; it is not substituted for discrete minimization on an arbitrary finite graph.

6 Numerical Verification and Illustrations

6.1 Numerical Methodology and Synthetic Environments

We evaluate architecture regret in synthetic, exogenously evolving Gaussian environments. The common-rate OU process in Eq. (50) is sampled exactly: for a step Δ, the autoregressive coefficient is $e^{-\Delta/T}$ and the innovation covariance is $(1 - e^{-2\Delta/T})\Sigma_S$. For affine decisions $x^\star = K\theta + k_0$, $K = Q^{-1}B$, we evaluate $R(\mathcal{A}) = \frac{1}{2}\mathbb{E}\|x^\star - u^\star_{\mathcal{A}}\|_Q^2$ both from conditional covariance and, where practical, from independently sampled quadratic losses. Error bars are 95% normal confidence intervals.

The constrained policy calculation matches the information structure in each experiment. Experiment 1 permits non-diagonal Q but uses common delayed global information, so the joint vector conditional expectation is exactly team-optimal. Experiment 2 uses the closed-form local rule derived from the coupled normal equations in Corollary 1. Experiments 3 and 4 evaluate the scalar regret expressions directly. Experiments 5 and 6 use $Q = I$, so agentwise Gaussian conditional expectations are team-optimal. No experiment substitutes componentwise conditional expectation for the solution of a heterogeneous-information team with off-diagonal Q.

Table 2 summarizes the reference run. Fixed seeds, raw trials, processed tables, covariance matrices, graph realizations, package versions, and condition diagnostics are archived in the accompanying reproduction package. The largest covariance condition number in the finite-graph experiments was 152.4, and every constructed covariance passed a positive-semidefinite eigenvalue check.

The experiments provide numerical verification and finite-graph illustrations within the assumed common-rate Gaussian-affine model. They are not tests of robustness to temporal-rate heterogeneity, nonseparable covariance, non-Gaussian evolution, nonlinear decision mappings, or mixed-age information architectures. Exact system matrices, graph seeds, conditioning thresholds, boundary conventions, radius grids, crossover estimators, and confidence-interval calculations are listed in the public reproduction documentation.

Table 2: Reference numerical settings. All state variances, local stiffnesses, and unswept coherence times are normalized to one.

Experiment	principal sweep	size	Monte Carlo samples
Temporal law	$\tau/T \in [0, 3]$	$n = 8, 20, 32, 50$	20,000/point
Local–global	$\gamma/q \in [0, 10]$	$n = 5, 10, 25, 100$	30,000/setting
Radius curves	$r/\ell_c \in [0, 2]$	three regimes	50,000/point
Radius phase	$vT/\ell_c \in [0.1, 20]$	72×68 grid	covariance
Decision relevance	$\ell_c = 0.7, 2, 6$	64-node ring	12,000/check
General graphs	path, ring, grid, geometric	64 nodes	covariance

6.2 Temporal Scaling and the Local–Global Crossover

We first tested Eq. (56) on four materially different systems: dimensions ranged from 8 to 50; Σ_S included independent, Toeplitz, graph-Laplacian, and geometric-graph covariances; and Q and B ranged from identity/dense to coupled/local operators. Across 100 parameter points, the Monte Carlo curves in Fig. 1 collapse onto $1 - e^{-2\tau/T}$. The largest absolute normalized deviation was 0.0155 and the root-mean-square deviation was 0.00471; the largest discrepancy between the empirical and direct covariance evaluations was likewise 0.0155. Confidence intervals are generally smaller than the plotted markers except at the largest dimensions and delays.

We next used the shared-resource model $Q = qI + \gamma \mathbf{1}\mathbf{1}^\top$, $q = 1$, with independent OU components. Figure 2 compares the empirical crossover with the exact finite-n threshold in Eq. (72). Over 36 $(n, \gamma/q)$ settings, the mean absolute boundary error was 0.00111 and the maximum was 0.00557. Finite size matters most under strong coupling: at $n = 5$ and $\gamma/q = 10$, the exact threshold is 0.109 below the large-system limit. As n grows, the measured boundary approaches $\frac{1}{2}\log(1 + \gamma/q)$, confirming that stronger decision coupling increases the amount of global staleness worth tolerating.

6.3 Emergence of an Optimal Coordination Radius

To expose the full tradeoff, we used $\eta_S(r) = \eta_0 e^{-2r/\ell_c}$, $\tau(r) = r/v$, and decomposed regret into the terms $R_T(r)$ and $R_S(r)$ of Eq. (122). Figure 3 deliberately selects local, interior, and boundary/global regimes on $0 \le r/\ell_c \le 2$. The grid minimizers were 0, 0.626, and 2.000, respectively; the analytical capped optima were 0, 0.62638, and 2.000. In the interior case the numerical radius error was 3.81×10^{-4} and $|m_S(\hat{r}^\star) - m_T(\hat{r}^\star)| = 4.77 \times 10^{-4}$, directly testing Eq. (129). Independent

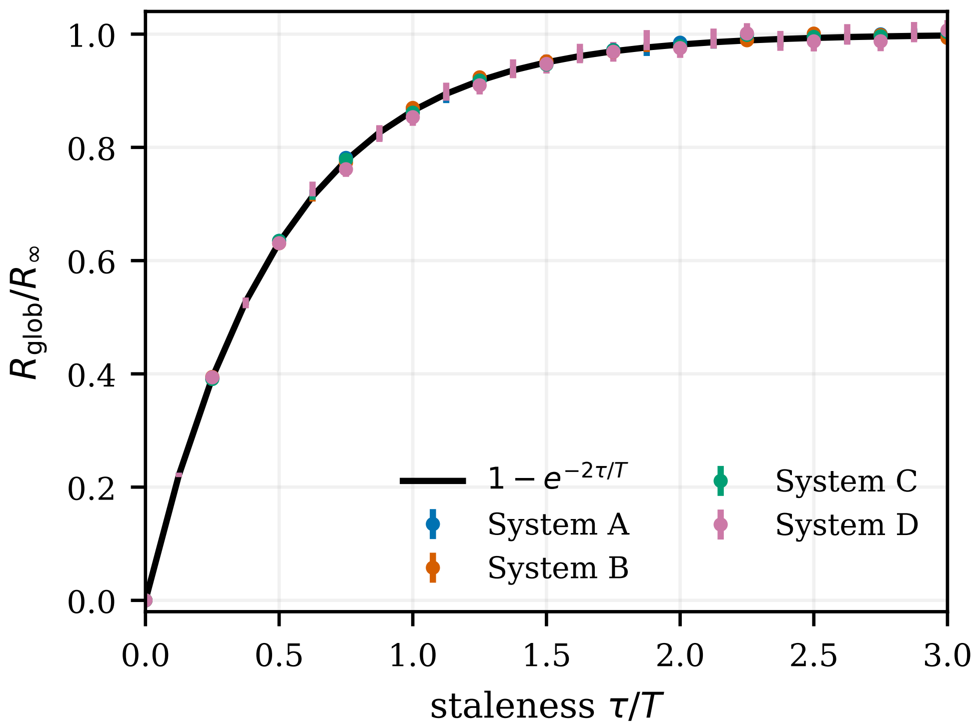

Figure 1: Temporal scaling collapse. Markers are Monte Carlo estimates for four systems with different dimensions, spatial covariances, objective matrices, and decision operators; bars are 95% confidence intervals. The solid curve is the exact law $1 - e^{-2\tau/T}$. Normalization by R_∞ removes all system-specific scale.

innovation sampling tracked the covariance curves with a worst-case absolute deviation of 0.020 over all radii and all three regimes.

6.4 Scaling Laws and Regime Map

We swept 4,896 points over $vT/\ell_c \in [0.1, 20]$ and $\ell_s/\ell_c \in [0.05, 5]$. At each point we minimized Eq. (103) on a 2,501-point radius grid independently of the closed form. Figure 4 shows the predicted zero-to-positive-radius threshold at $vT = 2\ell_s$ and the quantitative agreement with Eq. (159): mean absolute radius error 2.25×10^{-4}, maximum error 8.00×10^{-4}, against a maximum grid spacing 1.60×10^{-3}. Five grid points immediately adjacent to the transition were classified as zero rather than positive because their closed-form optima fell below one grid step; there were no discrepancies away from that resolution band. Separate one-dimensional numerical sweeps were monotone in all four predicted directions: $r^\star$ increased with T, v, and ℓ_c, and decreased with ℓ_s.

6.5 State Predictability versus Decision-Relevant Predictability

We next held a 64-node ring, its Laplacian covariance, $T = 6$, and the latency law fixed, changing only the normalized decision-sensitivity operator $K_{ij} \propto e^{-d_G(i,j)/\ell_c}$. Exact Gaussian conditioning gives the curves in Fig. 5. Increasing ℓ_c from 0.7 to 2 to 6 increased $\eta_S(0)$ from 0.0460 to 0.258 to 0.572 and moved the discrete optimum from 1 to 2 to 5 hops, even though the raw state correlation

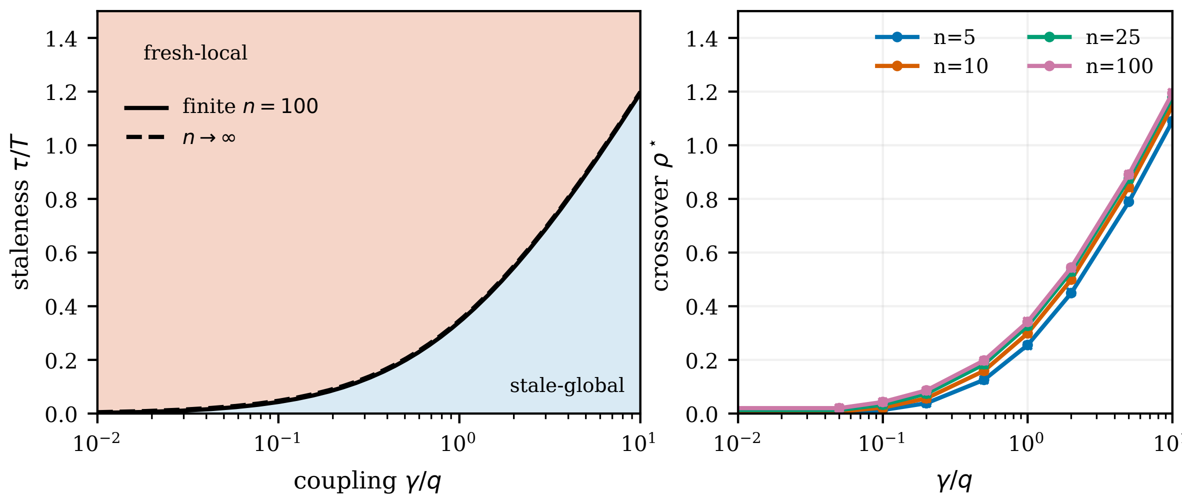

Figure 2: Fresh-local versus stale-global crossover. Left: winning architecture for $n = 100$; the solid finite-n boundary and dashed large-n boundary are Eqs. (72) and (74). Right: covariance boundaries (lines) and Monte Carlo estimates (crosses) for four system sizes. Stronger coupling makes older global information worth tolerating.

was identical. Monte Carlo conditioning checks differed from the exact $\eta_S(r)$ by at most 3.12×10^{-4}. Thus the same stochastic environment supports substantially different architectures depending on which spatial modes affect the desired decision.

6.6 Beyond the Canonical Line Model

Finally, we considered 64-node path, ring, two-dimensional grid, and random geometric graphs. For each graph we formed $\Sigma_S = (\kappa_s^2 I + L_G)^{-\nu}$, normalized its marginal variance, and computed every $\eta_S(r)$ directly from the Gaussian conditional covariance of the desired action given the radius-r neighborhood. No exponential fit or canonical line formula was used. Under the common reference setting, Fig. 6 gives optima of 2, 2, 3, and 2 hops for the path, ring, grid, and geometric graph, respectively.

Across all four graph classes, increasing T from 1.5 to 14 changed the optimal radius from 1 to 3 hops on the path and ring, from 1 to 4 on the grid, and from 1 to 3 on the geometric graph. Increasing the decision range from 0.7 to 6 moved the corresponding endpoints from $(0, 0, 1, 1)$ to $(5, 5, 5, 3)$. Increasing per-hop latency weakly decreased every optimum. Spatial smoothness was increased by decreasing κ_s (which emphasizes low graph frequencies after variance normalization); the optimal radii weakly decreased, as broader measurements became more redundant. Because radius is discrete, these stepwise changes should be interpreted using one-hop regret differences—the discrete analogue of marginal balance—rather than the continuous first-order condition in Eq. (129).

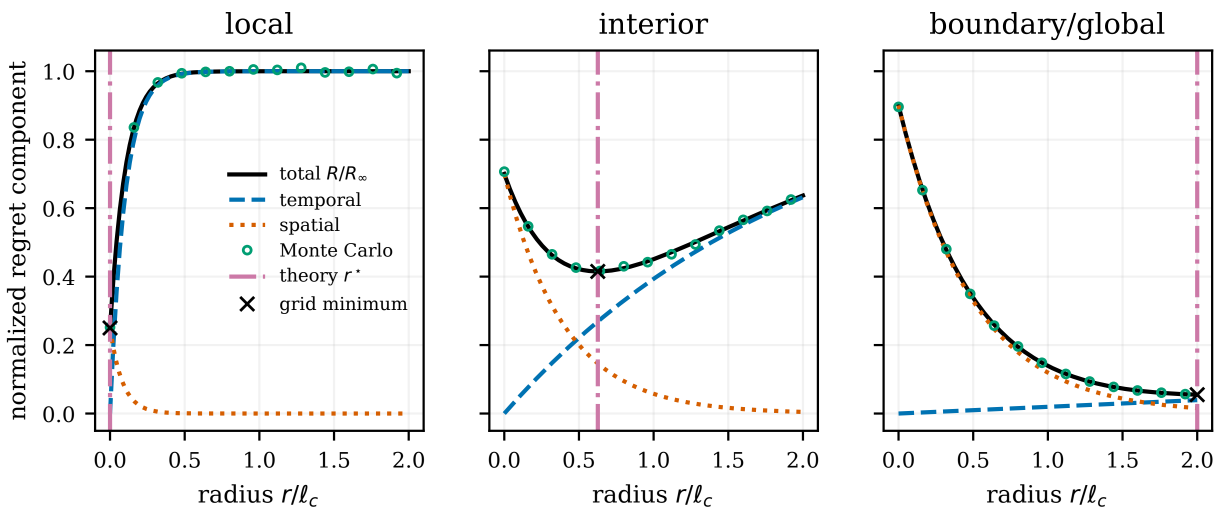

Figure 3: Emergence of local, interior, and boundary/global optima under $\eta_S(r) = \eta_0 e^{-2r/\ell_c}$ and $\tau(r) = r/v$. Curves show total regret and its temporal and residual-spatial terms from Eq. (122); open circles are independent Monte Carlo estimates. Dash-dotted lines and crosses mark the analytical and grid minimizers.

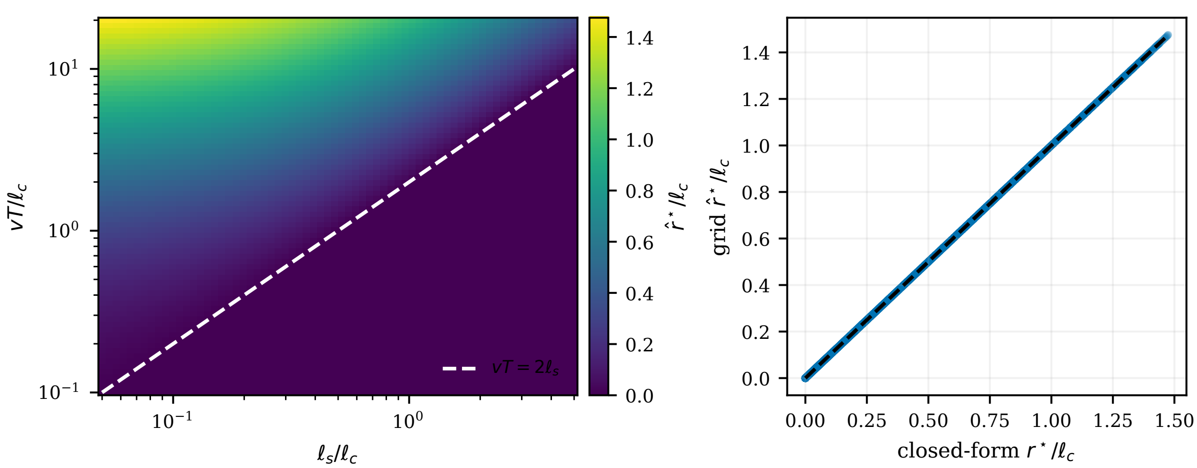

Figure 4: Dimensionless optimal-radius law. Left: numerical $r^\star/\ell_c$ over the $(\ell_s/\ell_c, vT/\ell_c)$ plane; the dashed line is the predicted transition $vT = 2\ell_s$. Right: grid minimizers versus the closed form in Eq. (159); the dashed line is equality.

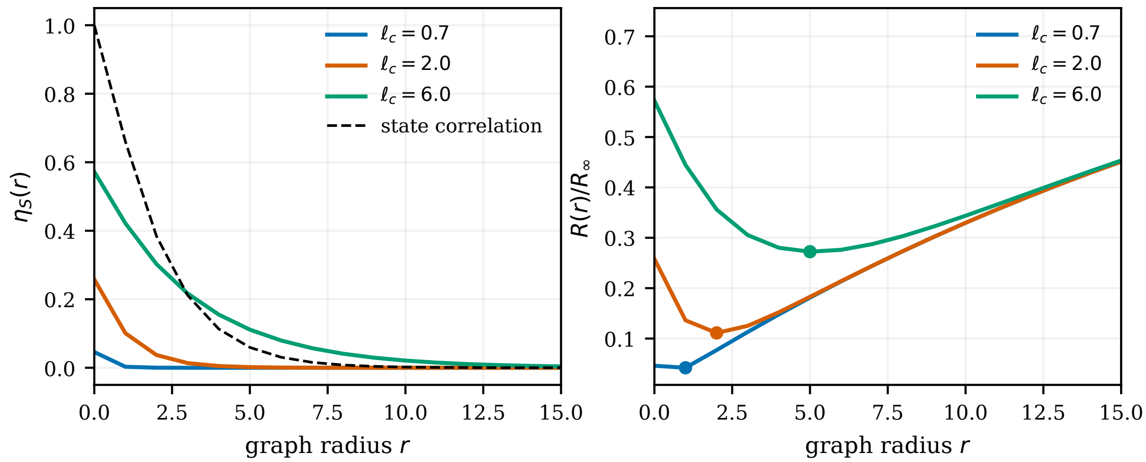

Figure 5: Decision relevance with the graph and state covariance held fixed. Left: exact Gaussian spatial omission for three normalized decision operators; the dashed raw state correlation is identical in every case. Right: composed regret with the minimizing radius marked. Changing only the decision-relevance range moves the optimum from 1 to 5 hops.

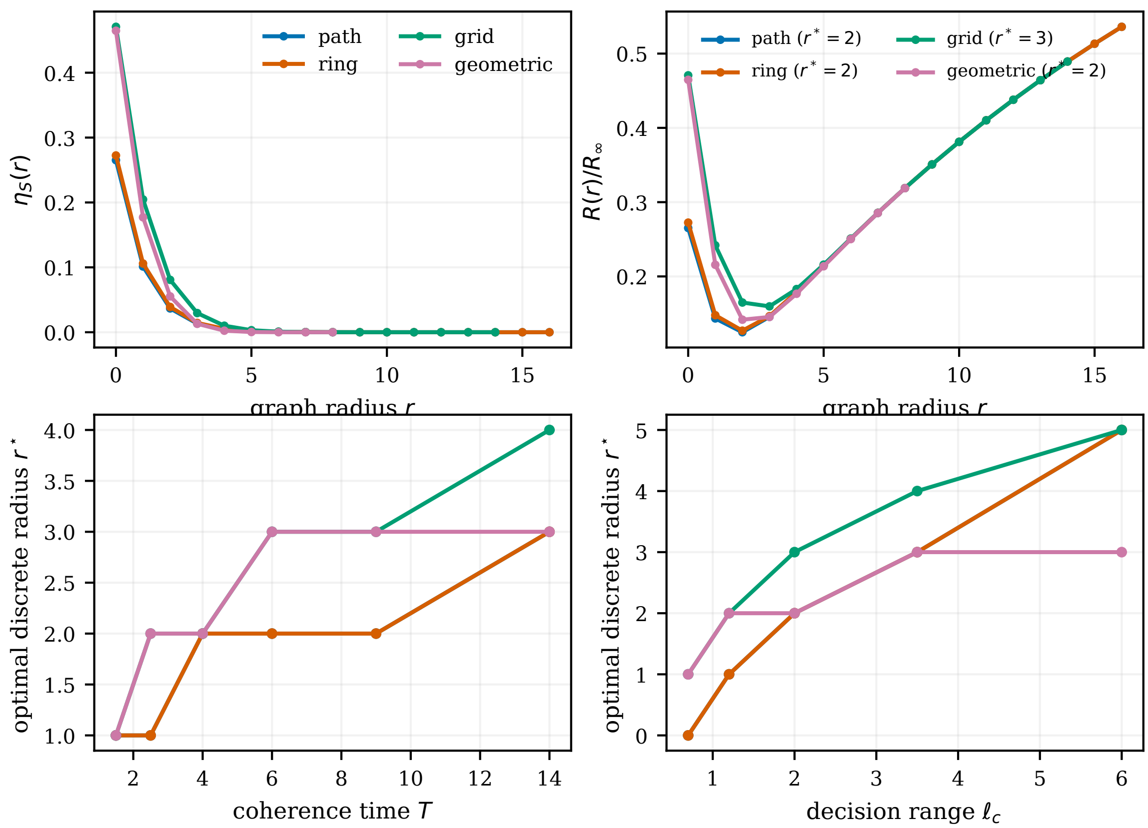

Figure 6: Finite noncanonical graphs. Top: $\eta_S(r)$ computed from actual Gaussian conditional covariances and the resulting regret under one common normalized setting. Bottom: discrete optimal radius versus coherence time and decision-relevance range. The canonical line closed form is not used; the stepwise shifts are the discrete analogue of marginal balance.

7 Related Work

7.1 Static Team Decision Theory

Radner established foundational optimality conditions for static teams, in which decision makers share a common objective but act on different information [13]. Under appropriate convexity and regularity conditions, the resulting stationarity conditions are sufficient for global team optimality; in the quadratic Gaussian specialization, the optimal decision rules are affine [13, 14]. In the deterministic-Hessian quadratic model used here, these fixed-information optimality conditions admit the equivalent Hilbert-space formulation developed in Section 2.4: for a fixed information architecture $\mathcal{A}$, the team-optimal policy is the Q-orthogonal projection of the full-information action onto the closed subspace $\mathcal{S}(\mathcal{A})$ of implementable policies. This projection identity is a specialization of the classical static-team framework to our quadratic model, rather than the literal statement of Radner's theorem. Krainak, Speyer, and Marcus subsequently relaxed the sufficient conditions associated with Radner's stationarity result and extended the theory to an exponential-quadratic criterion with jointly Gaussian state and observations [23]. In a companion paper, they formulated affine linear-quadratic-Gaussian and linear-exponential-Gaussian team problems as constrained parameter optimizations and represented optimal team gains as projections of the corresponding centralized gains [18].

Classical team theory also treated information itself as an organizational design object. Marschak and Radner analyzed the value of information, delayed information and the tradeoff between timeliness and completeness, and formulated team organization in terms of optimal networks [14]. More recently, Summers, Li, and Kamgarpour formulated information-structure design as the selection of additional measurements or communication links jointly with the corresponding team decision strategies [15]. Afshari and Mahajan characterized quadratic Gaussian static teams whose observations decompose into information common to all agents and agent-specific local information [24]. They subsequently applied this decomposition to team-optimal decentralized filtering over communication graphs with delayed information sharing, where sufficiently old measurements constitute common information while more recent measurements remain local [25]. Rusmevichientong and Van Roy studied large teams in which each agent observes the local cost structure only within a graph radius, using centralized performance as a benchmark [16]. Gamarnik, Goldberg, and Weber studied decentralized decisions based on local information in random decision networks and related their near-optimality to a correlation-decay, or long-range-independence, property [26].

Relative to these works, the distinctive object studied here is a structured family of information architectures in which spatial scope and temporal freshness are coupled through the scope–latency relation $\tau(r)$. Delayed information and timeliness–completeness tradeoffs are already present in classical team theory [14], while prior information-structure-design work selects additional information links [15] and prior local-information models study performance as a function of information radius [16]. Here, instead, each spatial scope induces an acquisition delay, and the resulting architecture is valued through its prediction error for the full-information decision. This coupling yields the freshness–locality crossover and coordination-scale laws developed in Sections 3 and 5.

7.2 Distributed Optimization, Communication Constraints, and Scope

Distributed optimization studies how agents compute a common solution using local computation and message exchange. Classical asynchronous gradient methods allow delayed interprocessor communication while retaining convergence guarantees under suitable boundedness conditions [27], while distributed subgradient methods establish convergence when agents exchange information locally over time-varying network topologies [3]. In these canonical formulations, the communication process constrains an iterative computation of an optimizer. Our setting separates that computational question from information architecture: the full-information decision at each epoch is the benchmark, and the object of design is the information available when that decision must be made.

Networked and decentralized control place communication restrictions directly in the feedback loop, including sampling, packet loss, communication topology, and delay [28]. For spatially invariant plants, early work showed that optimal controllers can exhibit inherent spatial localization [5], while subsequent work imposed explicit finite-speed communication structure. Voulgaris, Bianchini, and Bamieh studied H_2 control under delayed communication requirements [6]; Bamieh and Voulgaris formulated distance-dependent information propagation through funnel causality [7]; and Fardad and Jovanović developed state-space models for distributed controllers with finite communication speed [8]. These works are important antecedents to the present use of spatial scope and propagation time, but they optimize dynamic feedback controllers for controlled plants rather than information-constrained stage decisions in an exogenous environment.

Communication structure has also been treated as a design variable. Langbort and Gupta study interconnection topology when communication is assigned a topology-dependent cost [29], and Matni co-designs communication-delay structure and H_2 control using regularization [30]. In wireless consensus, Vanka, Gupta, and Haenggi explicitly trade communication radius against MAC/interference-induced delay and show that the preferred connectivity depends on network geometry [11]. Most directly, Ballotta, Jovanović, and Schenato parameterize control architectures by communication scope: each agent receives measurements from nodes within a chosen number of communication hops, with an architecture-dependent delay that increases with that scope. After optimizing the feedback gains for each scope, they show that sparse architectures can outperform all-to-all feedback [9]. Ballotta and Gupta obtain an analogous conclusion when maximizing consensus convergence speed under hop-dependent communication delay [10]. These results establish that scope-dependent delay can itself induce a sparse or intermediate optimal control architecture.

The distinction of the present paper is therefore *not* the existence of a scope–latency tradeoff, architecture-dependent delay, or a sparse/intermediate optimum. Our setting is an exogenous-state static team. The current full-information action is the benchmark, and an architecture is valued by the prediction error it induces for that action. This yields the decision-relevant spatial omission function $\eta_S(r)$, temporal unpredictability $\eta_T(\tau)$, their exact common-rate composition law, and the fresh-local/stale-global and coordination-radius laws derived above. Ballotta, Arbelaiz, Gupta, Schenato, and Jovanović study a different but complementary question for spatially invariant dynamic plants: how delayed measurements from different spatial locations alter the spatial locality of the optimal feedback kernel [12]. Their results operate through closed-loop plant dynamics and controller localization, whereas ours operate through prediction of an exogenously varying full-information decision.

7.3 Freshness and Age of Information

Age of Information (AoI) formalizes the timeliness of status information through the age of the freshest received update, $\Delta(t) = t - u(t)$. Seminal work showed that minimizing AoI is distinct from maximizing throughput or minimizing packet delay and initiated a large literature on update generation, queueing, and scheduling [2,31]. Subsequent work generalized linear age to application-specific nonlinear age penalties [32]. Connections to remote estimation make the distinction between age and task loss explicit: for a Gauss–Markov Ornstein–Uhlenbeck process, estimation error under signal-agnostic sampling is a nonlinear function of age, whereas signal-aware sampling can exploit the realized estimation error and outperform age-optimal policies [33].

More recent work has pushed freshness metrics toward task-aware objectives. Shisher et al.

showed that remote-inference loss can be nonmonotonic in age and designed communication policies around inference performance [34]; subsequent work considers correlated sources for which inference penalties depend jointly on multiple sources' ages [35]. Li et al. go further toward decision-aware freshness by jointly optimizing sampling and remote decision making under stale observations [36]. Our problem is complementary but distinct. Rather than optimizing when or which updates to transmit to a fixed remote receiver, we choose the information architecture available to a decentralized team—what is observed, from what spatial scope, and at what age. Because broader scope itself incurs greater acquisition delay, spatial omission and temporal staleness are coupled by the architecture. This leads to the fresh-local/stale-global crossover and optimal coordination-scale results developed here.

7.4 Centralization, Delegation, and Organizational Economics

A closely related organizational-economics literature asks how dispersed information, communication constraints, and decision rights determine organizational form. Aoki compares hierarchical and horizontal information structures when local conditions are uncertain and central monitoring or response is limited [37]. In the information-processing tradition, Radner studies organizations of limited-capacity processors trading off processing resources against decision delay [38]; Bolton and Dewatripont derive communication networks from processing and communication costs [39]; and Van Zandt studies real-time decentralized processing in which computational delay constrains the use of recent information [40]. Dessein and Santos likewise study adaptation to local information versus coordination under imperfect communication [41].

A complementary delegation literature makes decision rights endogenous when information and objectives are dispersed. Aghion and Tirole distinguish formal from real authority [42], while Dessein studies delegation as an alternative to strategic communication [43]. Most closely related to our centralization framing, Alonso, Dessein, and Matouschek compare centralized and decentralized coordination when managers privately observe local conditions [4]. Our model deliberately abstracts from incentive conflicts and endogenous information acquisition: agents share a common objective and the environment evolves exogenously. The tradeoff studied here is spatio-temporal: broader synchronized snapshots can improve coordination but arrive older. This yields a pure-benchmark crossover and, in the canonical spatial model, an interior optimum within the synchronized-snapshot family.

7.5 Spatial Statistics and Graph Signal Processing

Spatial statistics provides the closest statistical antecedent to our spatial-predictability model. Kriging casts spatial prediction as an optimal linear prediction problem under spatial dependence [44]. For lattice data, Besag's conditional formulation represents spatial dependence through local Markov neighborhoods [45]; Lindgren, Rue, and Lindström later connect Matérn Gaussian fields to sparse Gaussian Markov random fields through stochastic partial differential equation representations [21]. Space–time statistics likewise treats separability as a modeling restriction rather than a necessity: Gneiting constructs broad classes of stationary nonseparable covariance functions [22]. These works provide the statistical context for our covariance-based spatial model and its separable/nonseparable distinction.

Graph signal processing extends analogous ideas to irregular domains. Shuman et al. use graph-Laplacian eigenvectors to define a graph Fourier domain and associate low graph frequencies with signals that vary slowly over the graph [20]. Graph-sampling theory asks when bandlimited graph signals can be exactly recovered from partial observations [46], while stationary graph signal processing gives spectral descriptions and Wiener-type estimators for random graph signals [47]. Our objective is different: we do not value accurate reconstruction of θ_t per se. A radius-r architecture is evaluated by its loss in reproducing the full-information decision $x_t^\star$, and increasing r also increases information age through $\tau(r)$. Thus spatial omission and temporal staleness are priced jointly, and graph-smooth modes matter only insofar as they are decision-relevant.

8 Discussion and Conclusions

What the theory establishes. The central constraint in networked decision making is that information breadth and freshness generally cannot be chosen independently. Nearby observations can be used quickly but may omit remote state needed for coordination; a broader view can improve coordination but loses predictive value while it is collected and disseminated. An information architecture records this tradeoff by specifying what each agent knows, from where, and at what age. Architecture regret then measures the decision-quality loss relative to the full-information decision after optimizing over every policy implementable under that architecture. This separates the value of an information pattern from the particular protocol or algorithm used to exploit it.

The fresh-local versus stale-global result characterizes the two endpoints. Fresh-local information is preferred once the temporal unpredictability of the delayed global history exceeds the fraction of decision-relevant variation unavailable locally. Under the common-rate Gauss–Markov specialization, this comparison depends on the dimensionless delay τ/T and the locally captured share η_{loc}. The closed form is useful because it separates two mechanisms: the rate at which coordinated information ages and the amount of the full-information decision already inferable from local observations. The shared-resource example makes the second mechanism explicit: stronger coupling raises the value of global coordination and therefore the delay worth tolerating.

Intermediate radii define an outer design problem within the synchronized-snapshot family. Increasing radius replaces a narrower, fresher snapshot with a broader, older one. The optimum within this non-nested family occurs where the proportional marginal gain in decision-relevant spatial information equals the marginal loss of freshness. In the canonical line model, the spatial correlation length ℓ_s, the decision-relevance length ℓ_c, and the temporal propagation length $L_T = vT$ determine the dimensionless radius and the transition $L_T > 2\ell_s$. The numerical experiments verify the temporal collapse and local–global crossover, the zero-to-positive-radius threshold, the predicted movement of the optimum, and the same marginal-balance behavior on finite noncanonical graphs. Figures 1–4 provide implementation and finite-sample verification, while Figs. 5–6 illustrate decision relevance and finite-graph behavior. All experiments remain within the common-rate Gaussian-affine model.

Normalizing by open-loop regret supports comparisons across the modeled systems. It separates the overall scale of the decision problem from the fraction of decision-relevant variation preserved by an architecture. Systems with different dimensions, covariances, and coupling matrices can therefore obey the same temporal law when they share the relevant normalized predictability. The canonical formulas are exact, interpretable instances of this comparison; the marginal-balance condition is the more general result when spatial omission must instead be computed or estimated from the actual graph.

Why decision-relevant predictability matters. Raw state predictability is not the correct design criterion. Architecture regret depends on predicting the full-information decision $x^\star$, not on reconstructing every component of θ. A remote state component can be hard to predict but architecturally unimportant when it has little influence on the desired action. Conversely, a small residual uncertainty can matter when the decision-sensitivity operator K strongly amplifies that mode. The relevant predictability therefore combines environmental statistics, decision sensitivity and coupling, information geometry, and freshness. Holding the state covariance fixed while varying K in the numerical section demonstrates that identical raw predictability can support substantially different optimal coordination radii.

This distinction also separates physical communication geometry from decision geometry. Graph distance determines which observations can be acquired and their age on arrival; Q, B, and K determine whether those observations change the desired joint action. Two nearby nodes may be weakly coupled, while distant nodes may interact strongly through a shared constraint or common decision mode. An architecture based only on physical distance can therefore communicate extensively about irrelevant variation while omitting a remote mode with high decision value. Architecture design must use the communication graph and the decision structure together.

The framework suggests broader information-architecture questions involving topology, representation, and timing. The present results analyze scope and timing for a synchronized-snapshot family; nonuniform topology and compressed representation remain extensions.

Scope and deliberate limitations. The paper establishes an exogenous-state, static-team baseline. Environmental state may be correlated across space and time, but current actions neither alter its future law nor change what another agent later observes. Within this boundary, the fixed-architecture quadratic projection gives an exact benchmark for each fixed information pattern, and the paper's new results compare a structured scope–age family using that benchmark. The principal radius family uses synchronized snapshots. Under the common-rate Gaussian model, Lemma 2 makes current snapshots sufficient relative to observed histories for the affine current-decision prediction problem; the equivalence is not assumed more generally.

The exact projection formula uses quadratic costs. The explicit stale-global and composition laws additionally use affine full-information decisions and a common-rate Gaussian or Gauss–Markov process, while the canonical radius formula uses a separable homogeneous field and linear propagation delay. These assumptions produce interpretable identities and dimensionless scaling laws. The general lessons are broader than the canonical formula, but outside these assumptions one should work directly with decision-relevant prediction-error functions or suitable bounds rather than assume the same factorization. In particular, nonquadratic and discrete decisions require analytical tools beyond the Hilbert-space projection argument.

The exact projection characterization also assumes unconstrained Euclidean action spaces. Nonnegativity, box, simplex, capacity, or other coupled feasibility constraints generally replace the linear policy subspace by a convex or nonconvex feasible set and require a different analysis. The projection identities also assume a deterministic action Hessian Q; a state-dependent or random Hessian may require a random weighted geometry and, even under common information, need not reduce to the same ordinary conditional-expectation formula.

The temporally evolving environment is treated as a sequence of stagewise static teams: current actions neither alter later state, reshape another agent's information, nor constrain later actions. The fixed-epoch projection does not require stationarity; stationarity makes expected performance time-invariant, while ergodicity is additionally needed for sample-path long-run averages.

Finally, the common-rate assumption suppresses possible alignment between temporal coherence, spatial scale, and decision relevance. If long-range decision-relevant modes evolve more slowly than local modes, a scalar common-rate approximation may understate the value of broad delayed information; the reverse alignment may overstate it. There is no universal direction of bias without specifying the mode weights and the procedure used to select an effective coherence time. With heterogeneous temporal modes, the scalar composition law generally gives way to a joint spatio-temporal decision-prediction error.

Most important extensions. A first extension is to price information acquisition explicitly. The present model represents communication and sensing constraints mainly through the feasible scope–delay relationship. A richer problem would minimize $R(\mathcal{A}) + \lambda C(\mathcal{A})$, or impose a budget on $C(\mathcal{A})$, where cost may represent bandwidth, energy, sensing effort, or computation. Explicit cost naturally leads to sparse and nonuniform topologies: different agents need not use the same radius, and a small number of selected long-range links may capture decision-relevant modes more efficiently than a complete global view. When remote state matters mainly through a low-dimensional statistic, compressed summaries may likewise retain most of the coordination value while reducing latency.

A second extension is adaptation to unknown or nonstationary statistics. Temporal coherence, spatial predictability, and decision sensitivity may need to be estimated online, with the architecture changing on a slower timescale than the operational decisions. The central challenge is then to account for uncertainty in the estimated prediction-error functions while retaining a stable architecture-selection rule.

An important extension is action-dependent state evolution and dynamic teams. Queues depend on service and routing decisions, batteries on energy use, thermal states on computation, mobility on previous control, and controlled resources on earlier allocations. In such systems, actions alter future state and may also signal information to other agents. The present static-team theory deliberately establishes the exogenous-state baseline, while extending information-architecture design to settings in which actions alter future state and/or future information is an important next research direction. Witsenhausen's counterexample [17] explains why this is not a routine modification: once control and signaling interact, the fixed-subspace projection characterization generally no longer solves the team problem.

The present theory suggests the following design heuristic: expand information scope only while the marginal decision value of broader information exceeds the predictive value lost while obtaining it.

Code and data availability. The complete manuscript source, numerical code, configurations, raw and processed results, and figure-generation pipeline are available in the TINA reproduction repository [48]. The cited release records the exact submission version and immutable commit.

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